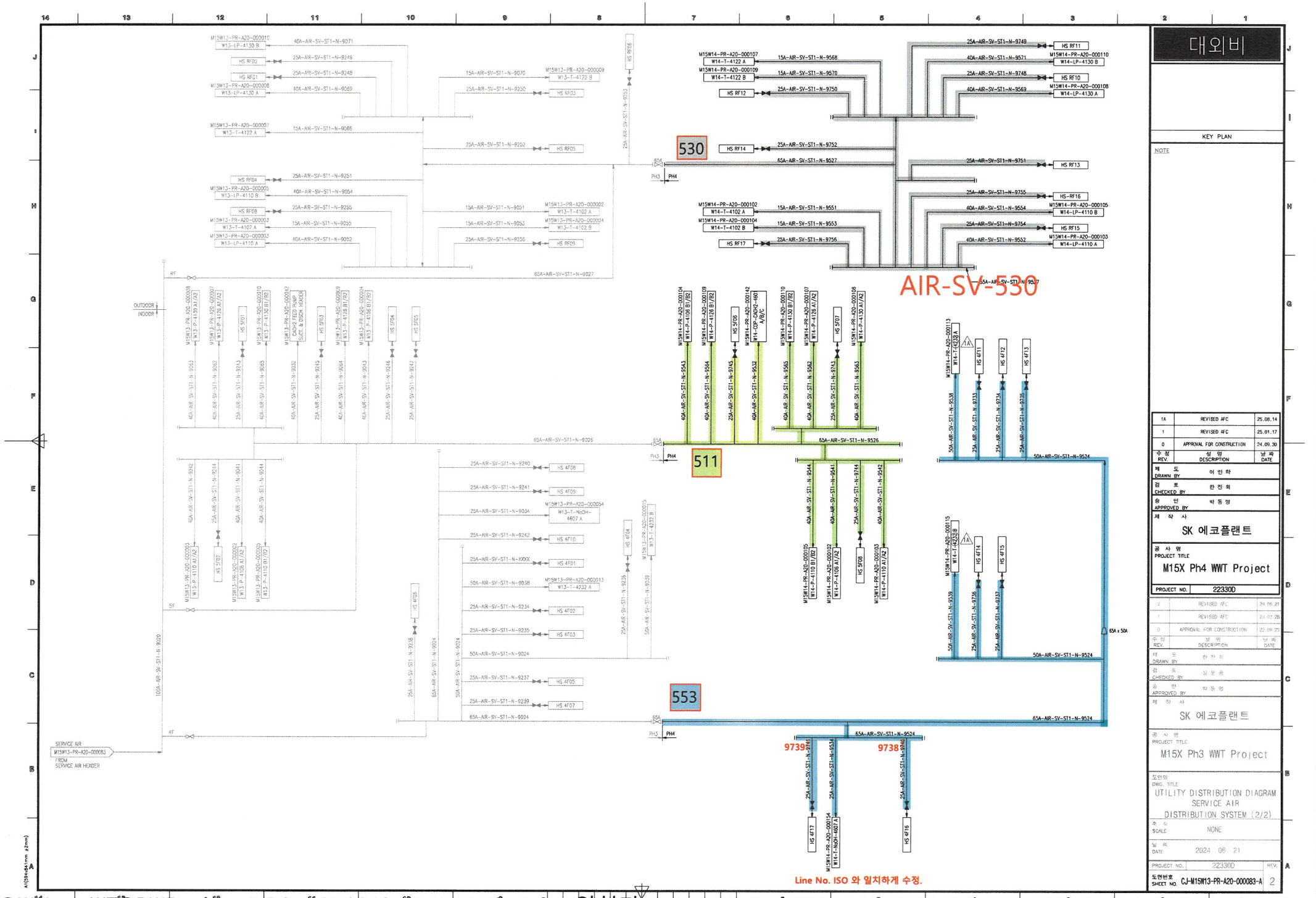
DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

AIR-SV-530

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	530	RFA	15A-AIR-SV-ST1-N-9551-01	1 OF 1	0	CJ-M15W14-PR-A20-000102	1A	
2	530	RFA	15A-AIR-SV-ST1-N-9553-01	1 OF 1	0	CJ-M15W14-PR-A20-000104	1A	
3	530	RFA	15A-AIR-SV-ST1-N-9568-01	1 OF 1	0	CJ-M15W14-PR-A20-000107	1A	
4	530	RFA	15A-AIR-SV-ST1-N-9570-01	1 OF 1	0	CJ-M15W14-PR-A20-000109	1A	
5	530	RFA	25A-AIR-SV-ST1-N-9755-01	1 OF 1	0	CJ-M15W13-PR-A20-000083-A	2	9255 > 9755 변경
6	530	RFA	25A-AIR-SV-ST1-N-9748-01	1 OF 1	0	CJ-M15W13-PR-A20-000083-A	2	
7	530	RFA	25A-AIR-SV-ST1-N-9749-01	1 OF 1	0	CJ-M15W13-PR-A20-000083-A	2	
8	530	RFA	25A-AIR-SV-ST1-N-9750-01	1 OF 1	0	CJ-M15W13-PR-A20-000083-A	2	
9	530	RFA	25A-AIR-SV-ST1-N-9751-01	1 OF 1	0	CJ-M15W13-PR-A20-000083-A	2	
10	530	RFA	25A-AIR-SV-ST1-N-9752-01	1 OF 1	0	CJ-M15W13-PR-A20-000083-A	2	
11	530	RFA	25A-AIR-SV-ST1-N-9754-01	1 OF 1	0	CJ-M15W13-PR-A20-000083-A	2	
12	530	RFA	25A-AIR-SV-ST1-N-9756-01	1 OF 1	0	CJ-M15W13-PR-A20-000083-A	2	
13	530	RFA	40A-AIR-SV-ST1-N-9552-01	1 OF 1	0	CJ-M15W14-PR-A20-000103	1A	
14	530	RFA	40A-AIR-SV-ST1-N-9554-01	1 OF 1	0	CJ-M15W14-PR-A20-000105	1A	
15	530	RFA	40A-AIR-SV-ST1-N-9569-01	1 OF 1	0	CJ-M15W14-PR-A20-000108	1A	
16	530	RFA	40A-AIR-SV-ST1-N-9571-01	1 OF 1	0	CJ-M15W14-PR-A20-000110	1A	
17	530	RFA	65A-AIR-SV-ST1-N-9527-01	1 OF 4	0	CJ-M15W13-PR-A20-000083-A	2	
18	530	RFA	65A-AIR-SV-ST1-N-9527-02	2 OF 4	0	CJ-M15W13-PR-A20-000083-A	2	
19	530	RFA	65A-AIR-SV-ST1-N-9527-03	3 OF 4	0	CJ-M15W13-PR-A20-000083-A	2	
20	530	RFA	65A-AIR-SV-ST1-N-9527-04	4 OF 4	0	CJ-M15W13-PR-A20-000083-A	2	



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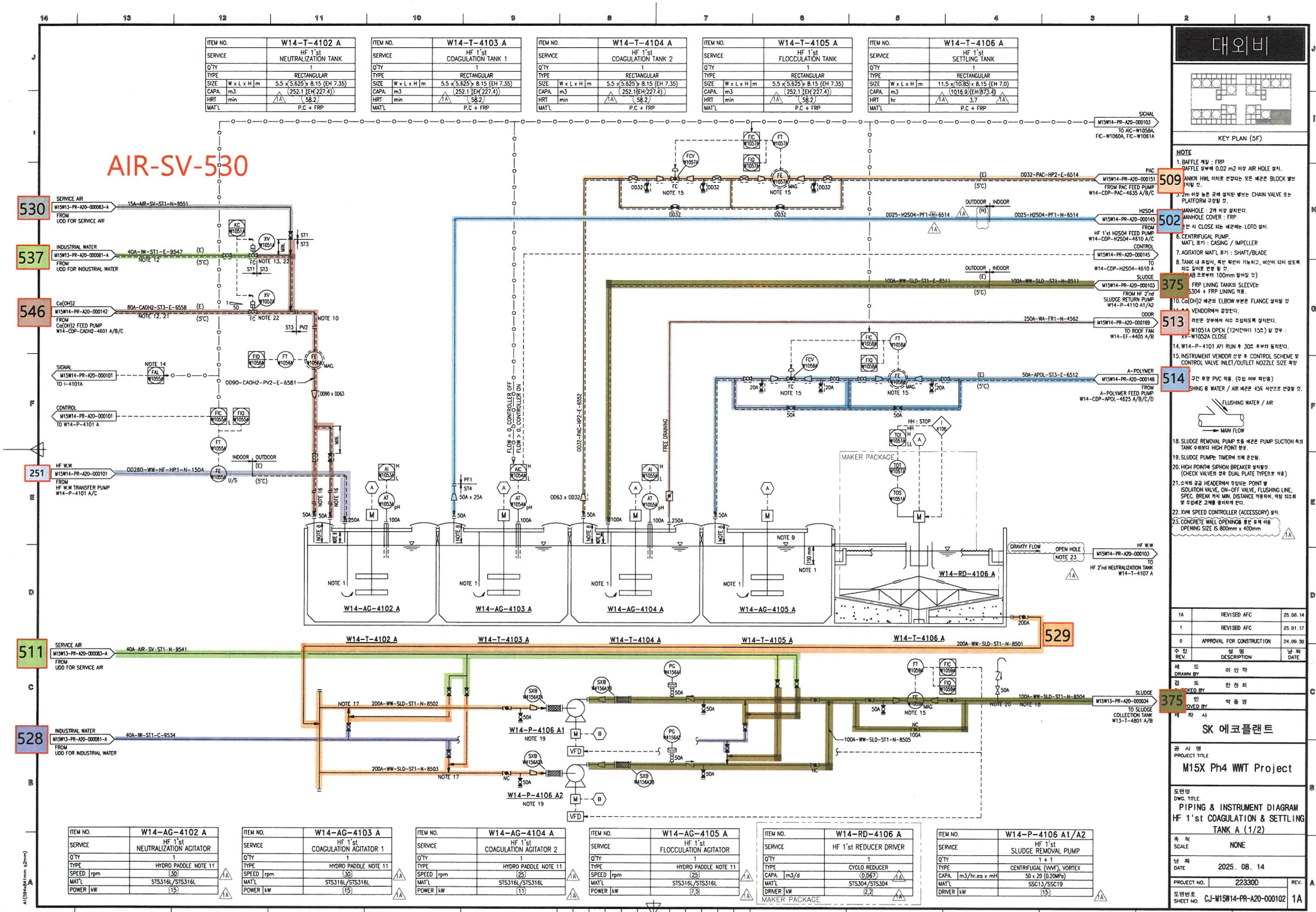
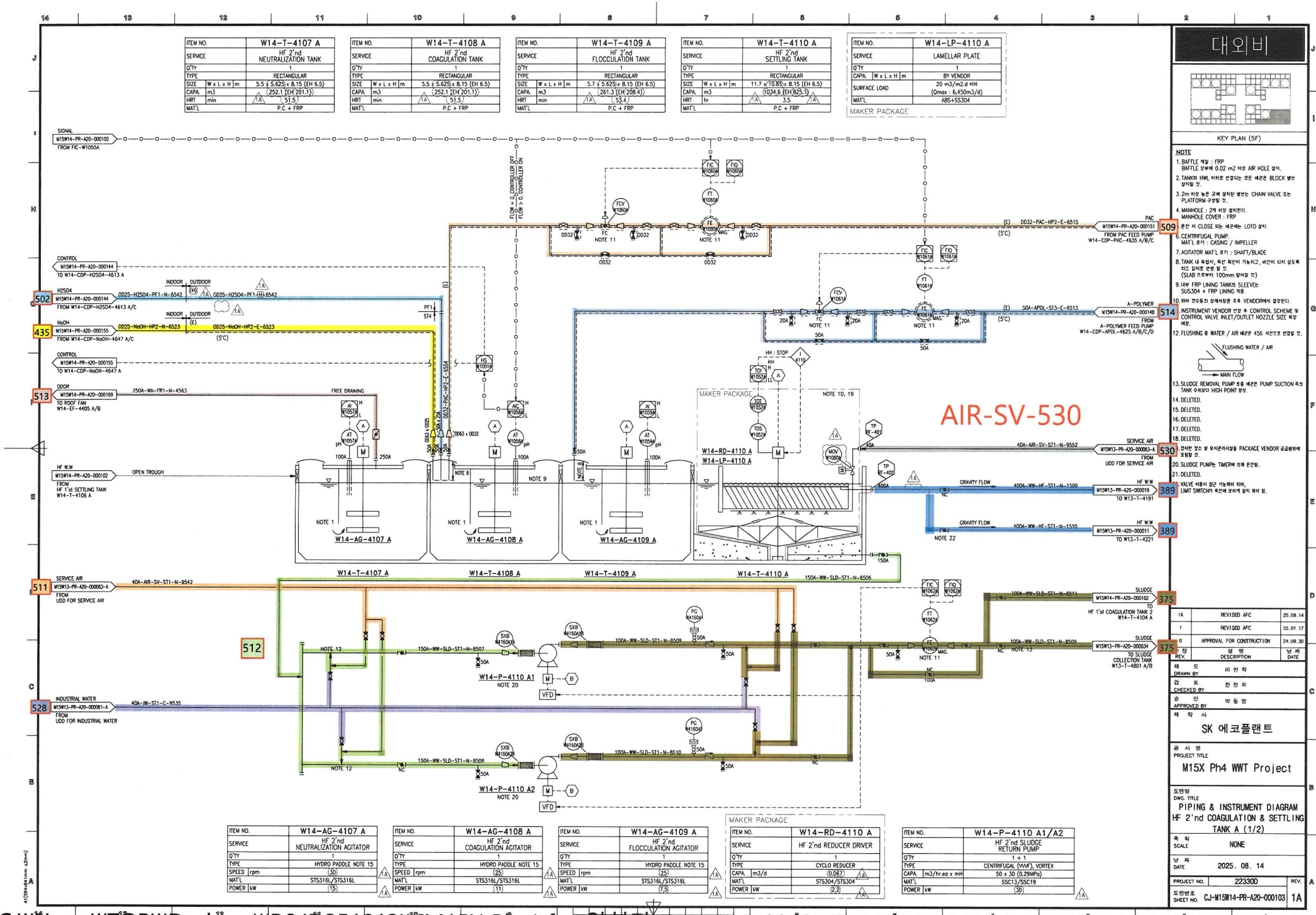
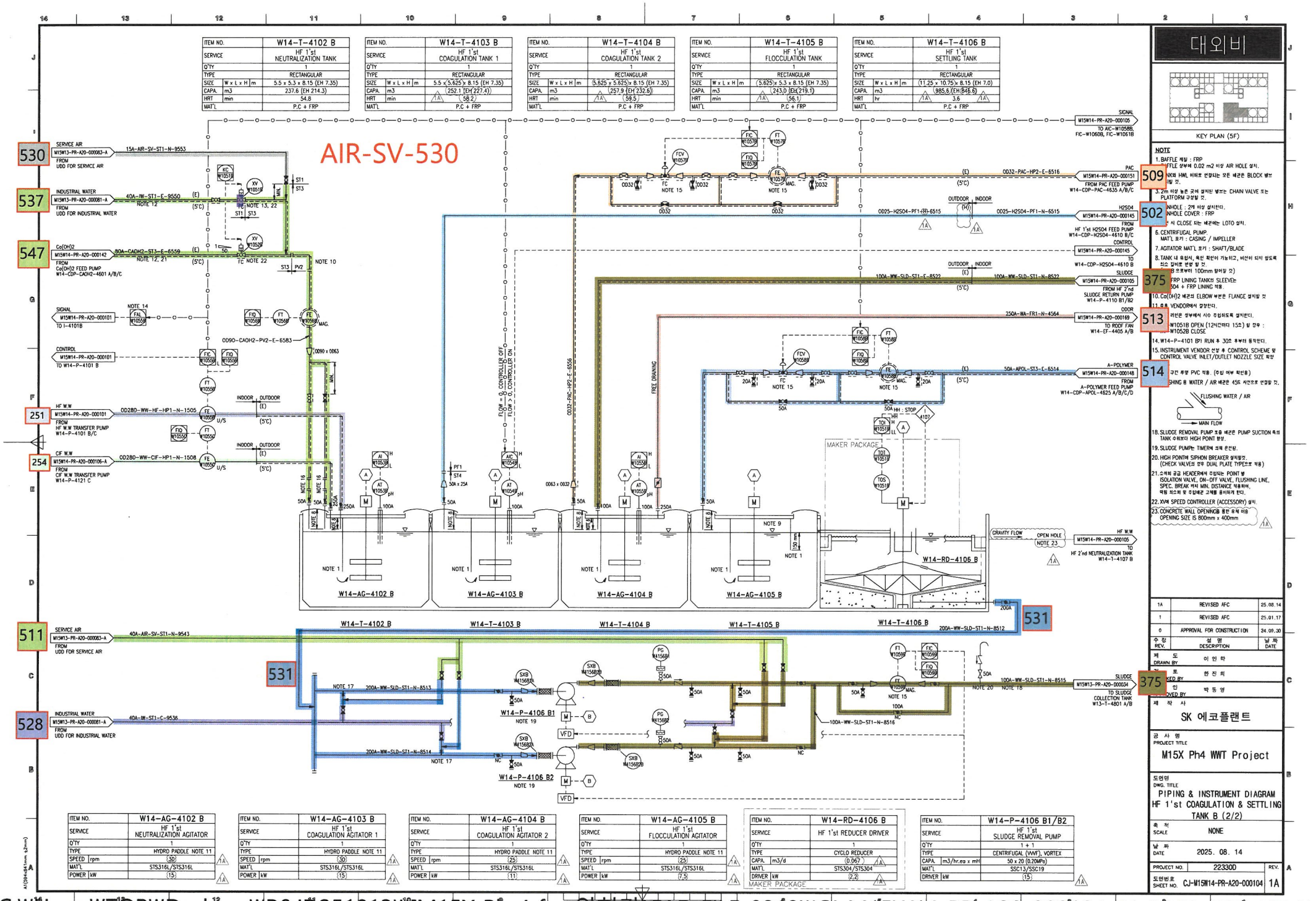

AIR-SV-530
대외비
KEY PLAN (5F)
W14-T-4102 A
HF 1'st NEUTRALIZATION TANK
W14-T-4103 A
HF 1'st COAGULATION TANK 1
W14-T-4104 A
HF 1'st COAGULATION TANK 2
W14-T-4105 A
HF 1'st FLOCCULATION TANK
W14-T-4106 A
HF 1'st SETTLING TANK
W14-AG-4102 A
W14-AG-4103 A
W14-AG-4104 A
W14-AG-4105 A
W14-RD-4106 A
W14-P-4106 A1
W14-P-4106 A2
MAKER PACKAGE
SK 에코플랜트
M15X Ph4 WWT Project
PIPING & INSTRUMENT DIAGRAM HF 1'st COAGULATION & SETTLING TANK A (1/2)
C:₩Users₩TOPWDesktop₩P&ID251013₩[M15X Ph 4 for 원어기]CAD FILE_0910₩CJ-M15W14-PR-A20-000102 R1 P&ID FOR HF 1's


대외비
AIR-SV-530
W14-T-4107 A
HF 2'nd NEUTRALIZATION TANK
W14-T-4108 A
HF 2'nd COAGULATION TANK
W14-T-4109 A
HF 2'nd FLOCCULATION TANK
W14-T-4110 A
HF 2'nd SETTLING TANK
W14-LP-4110 A
LAMELLAR PLATE
MAKER PACKAGE
KEY PLAN (5F)
NOTE
1. BAFFLE 재질 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
2. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
3. 2m 이상 높은 곳에 설치할 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
4. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
5. 운전 시 CLOSE 되는 배관에는 LOTO 설치.
6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
7. AGITATOR MAT'L 표기 : SHAFT/BLADE
8. TANK 내 유입부, 확산 확인이 가능하고, 버킷이 되지 않도록 최소 깊이로 반영 할 것.
(SLAB 으로부터 100mm 띄어질 것)
9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
10. 위에 것으들의 상세사항은 추후 VENDOR에서 결정한다.
INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정 예정.
12. FLUSHING 용 WATER / AIR 배관은 45도 사선으로 연결할 것.
13. SLUDGE REMOVAL PUMP 토출 배관은 PUMP SUCTION 측의 TANK 수위보다 HIGH POINT 형성.
14. DELETED.
15. DELETED.
16. DELETED.
17. DELETED.
18. DELETED.
19. 결빙한 장소 및 유지관리사항을 PACKAGE VENDOR 공급범위에 포함할 것.
20. SLUDGE PUMP는 TIMER에 의해 운전됨.
21. DELETED.
VALVE 제품이 접근 가능하여 하며, LIMIT SWITCH가 육안에 보이게 설치 해야 함.
1A REVISED AFC 25.08.14
1 REVISED AFC 25.01.17
0 APPROVAL FOR CONSTRUCTION 24.09.30
REV. 정 설 명 DESCRIPTION 날 짜 DATE
제 도 DRAWN BY 이 인 학
검 토 CHECKED BY 한 진 희
승 인 APPROVED BY 박 동 영
제 작 사
SK 에코플랜트
공 사 명 PROJECT TITLE
M15X Ph4 WWT Project
도면명 DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
HF 2'nd COAGULATION & SETTLING TANK A (1/2)
축 척 SCALE NONE
날 짜 DATE 2025. 08. 14
PROJECT NO. 22330D REV.
도면번호 SHEET NO. CJ-M15W14-PR-A20-000103 1A
W14-AG-4107 A
HF 2'nd NEUTRALIZATION AGITATOR
W14-AG-4108 A
HF 2'nd COAGULATION AGITATOR
W14-AG-4109 A
HF 2'nd FLOCCULATION AGITATOR
W14-RD-4110 A
HF 2'nd REDUCER DRIVER
W14-P-4110 A1/A2
HF 2'nd SLUDGE RETURN PUMP
SIGNAL M15W14-PR-A20-000102 FROM FIC-W1055A
CONTROL M15W14-PR-A20-000144 TO W14-CDP-H2SO4-4613 A
H2SO4 M15W14-PR-A20-000144 FROM W14-CDP-H2SO4-4613 A/C
NaOH M15W14-PR-A20-000155 FROM W14-CDP-NaOH-4647 A/C
CONTROL M15W14-PR-A20-000155 TO W14-CDP-NaOH-4647 A
ODOR M15W14-PR-A20-000169 TO ROOF FAN W14-EF-4405 A/B
HF W.W M15W14-PR-A20-000102 FROM HF 1'st SETTLING TANK W14-T-4106 A
SERVICE AIR M15W13-PR-A20-000083-A FROM UDD FOR SERVICE AIR
INDUSTRIAL WATER M15W13-PR-A20-000081-A FROM UDD FOR INDUSTRIAL WATER
PAC M15W14-PR-A20-000151 FROM PAC FEED PUMP W14-CDP-PAC-4635 A/B/C
A-POLYMER M15W14-PR-A20-000148 FROM A-POLYMER FEED PUMP W14-CDP-APOL-4625 A/B/C/D
SERVICE AIR M15W13-PR-A20-000083-A FROM UDD FOR SERVICE AIR
HF W.W M15W13-PR-A20-000019 TO W13-T-4191
HF W.W M15W13-PR-A20-000011 TO W13-T-4221
SLUDGE M15W14-PR-A20-000102 TO HF 1'st COAGULATION TANK 2 W14-T-4104 A
SLUDGE M15W13-PR-A20-000034 TO SLUDGE COLLECTION TANK W13-T-4801 A/B
W14-P-4110 A1
W14-P-4110 A2
VFD
OPEN TROUGH
FREE DRAINING
INDOOR
OUTDOOR
GRAVITY FLOW
FLUSHING WATER / AIR
MAIN FLOW
NOTE 1
NOTE 9
NOTE 10, 19
NOTE 11
NOTE 12
NOTE 13
NOTE 20
NOTE 22
C:₩Users₩T0PW0Desktop₩P&ID251013₩M15X Ph 4 for 외부기관 [읽기 전용]₩CAD FILE 0910₩CJ-M15W14-PR-A20-000103 R1 P&ID FOR HF 2'r



ITEM NO.	W14-AG-4102 B
SERVICE	HF 1'st NEUTRALIZATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 11
SPEED rpm	(30)
MAT'L	STS316L/STS316L
POWER kW	(15)

ITEM NO.	W14-AG-4103 B
SERVICE	HF 1'st COAGULATION AGITATOR 1
Q'TY	1
TYPE	HYDRO PADDLE NOTE 11
SPEED rpm	(30)
MAT'L	STS316L/STS316L
POWER kW	(15)

ITEM NO.	W14-AG-4104 B
SERVICE	HF 1'st COAGULATION AGITATOR 2
Q'TY	1
TYPE	HYDRO PADDLE NOTE 11
SPEED rpm	(25)
MAT'L	STS316L/STS316L
POWER kW	(11)

ITEM NO.	W14-AG-4105 B
SERVICE	HF 1'st FLOCCULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 11
SPEED rpm	(25)
MAT'L	STS316L/STS316L
POWER kW	(7.5)

ITEM NO.	W14-RD-4106 B
SERVICE	HF 1'st REDUCER DRIVER
Q'TY	1
TYPE	CYCLO REDUCER
CAPA. m3/d	(0.067)
MAT'L	STS304/STS304
DRIVER kW	(2.2)

MAKER PACKAGE

ITEM NO.	W14-P-4106 B1/B2
SERVICE	HF 1'st SLUDGE REMOVAL PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (VVF), VORTEX
CAPA. m3/hr.ea x mH	50 x 20 (0.20MPa)
MAT'L	SSC13/SSC19
DRIVER kW	(15)

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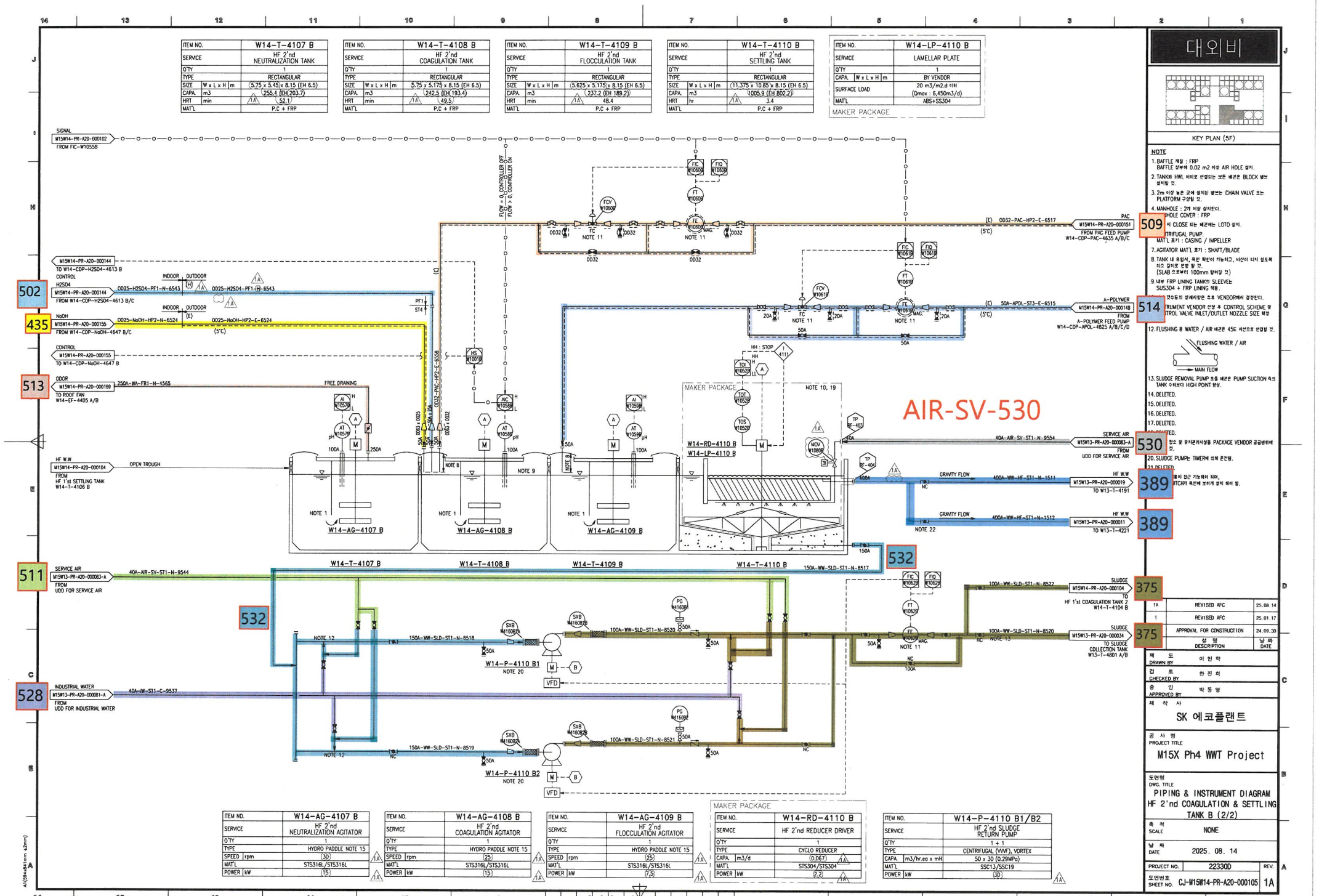

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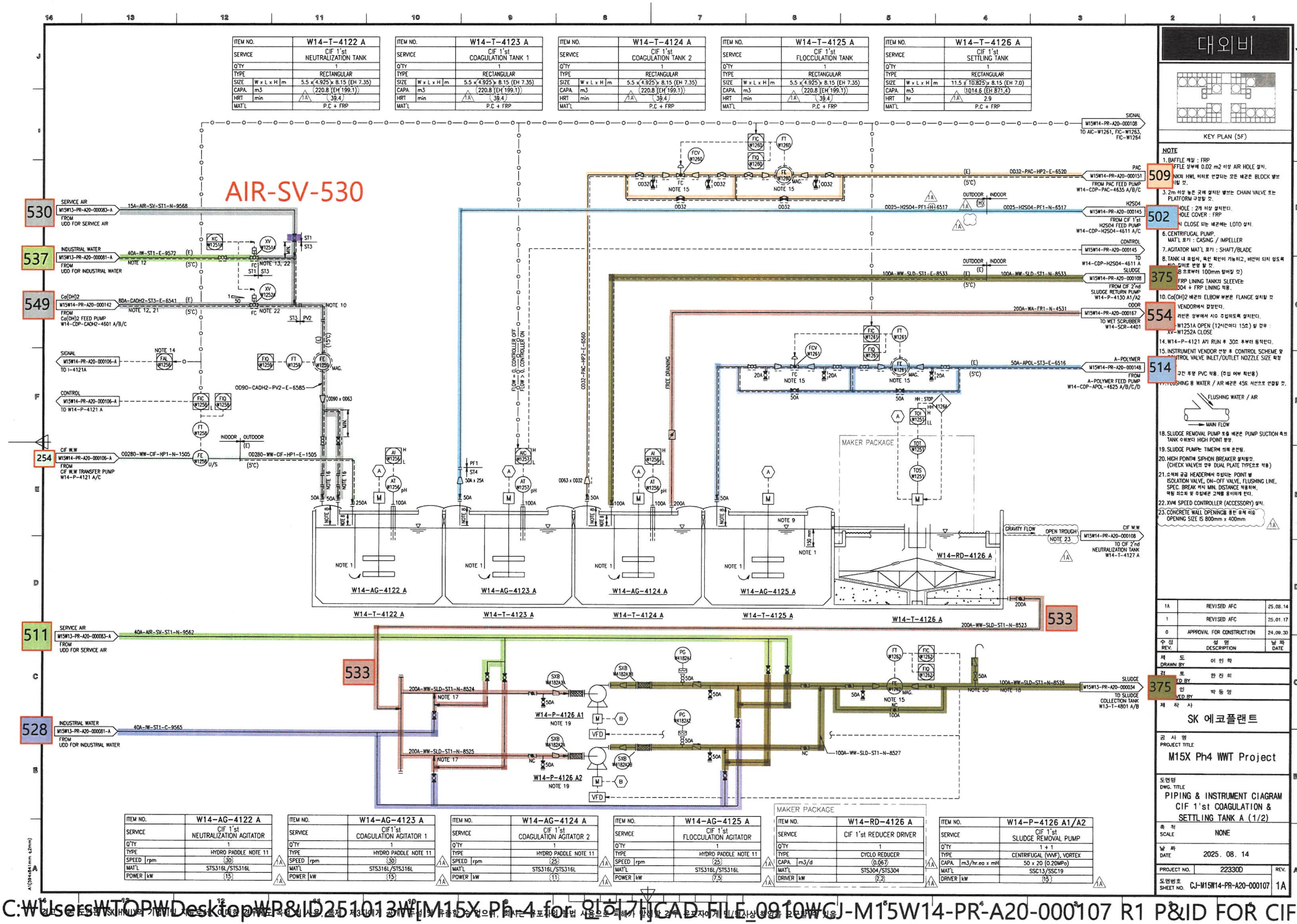

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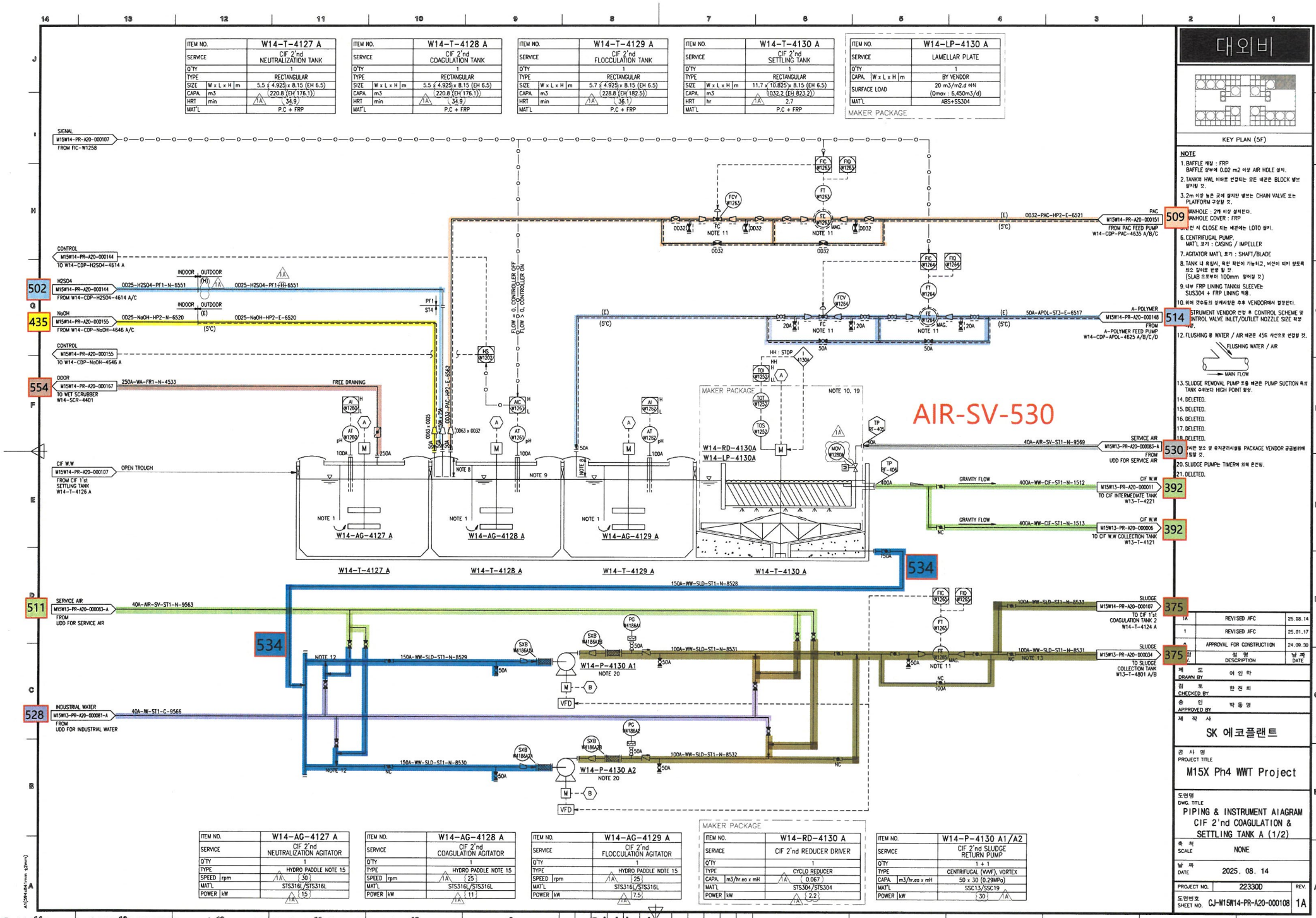
대외비
AIR-SV-530
W14-T-4127 A
CIF 2'nd NEUTRALIZATION TANK
W14-T-4128 A
CIF 2'nd COAGULATION TANK
W14-T-4129 A
CIF 2'nd FLOCCULATION TANK
W14-T-4130 A
CIF 2'nd SETTLING TANK
W14-LP-4130 A
LAMELLAR PLATE
MAKER PACKAGE
W14-AG-4127 A
CIF 2'nd NEUTRALIZATION AGITATOR
W14-AG-4128 A
CIF 2'nd COAGULATION AGITATOR
W14-AG-4129 A
CIF 2'nd FLOCCULATION AGITATOR
W14-RD-4130 A
CIF 2'nd REDUCER DRIVER
W14-P-4130 A1/A2
CIF 2'nd SLUDGE RETURN PUMP
W14-P-4130 A1
W14-P-4130 A2
KEY PLAN (5F)
NOTE
1. BAFFLE 재질 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
2. TANK와 WALL 이어로 연결되는 모든 배관은 BLOCK 넣브 설치할 것.
3. 2m 이상 높은 곳에 설치된 넣브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
4. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
5. 전 시 CLOSE 되는 배관에는 LOTO 설치.
6. CENTRIFUGAL PUMP,
MAT'L 표기 : CASING / IMPELLER
7. AGITATOR MAT'L 표기 : SHAFT/BLADE
8. TANK 내 유입시, 육안 확인이 가능하고, 바인이 더이 방도록 최소 넣이로 변경 할 것
(SLAB 으로부터 100mm 정어질 것)
9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 처용.
10. 이어 연수등의 상세사항은 추후 VENDOR에 결정한다.
11. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정 할것.
12. FLUSHING 용 WATER / AIR 배관은 45도 사선으로 변경할 것.
13. SLUDGE REMOVAL PUMP 로출 배관은 PUMP SUCTION 측의 TANK 수위보다 HIGH POINT 설정.
14. DELETED.
15. DELETED.
16. DELETED.
17. DELETED.
18. DELETED.
19. 사선은 장소 및 유지관리사정을 PACKAGE VENDOR 공급범위에 포함할 것.
20. SLUDGE PUMP는 TIMER에 의해 운전됨.
21. DELETED.
REVISED AFC 25.08.14
REVISED AFC 25.01.17
APPROVAL FOR CONSTRUCTION 24.09.30
제 도 DRAWN BY 이 인 학
검 토 CHECKED BY 한 전 희
승 인 APPROVED BY 박 동 영
제 작 사
SK 에코플랜트
공 사 명 PROJECT TITLE
M15X Ph4 WWT Project
도면명 DWG. TITLE
PIPING & INSTRUMENT AIAGRAM
CIF 2'nd COAGULATION & SETTLING TANK A (1/2)
축 척 SCALE NONE
날 짜 DATE 2025. 08. 14
PROJECT NO. 22330D
도면번호 SHEET NO. CJ-M15W14-PR-A20-000108
REV. 1A
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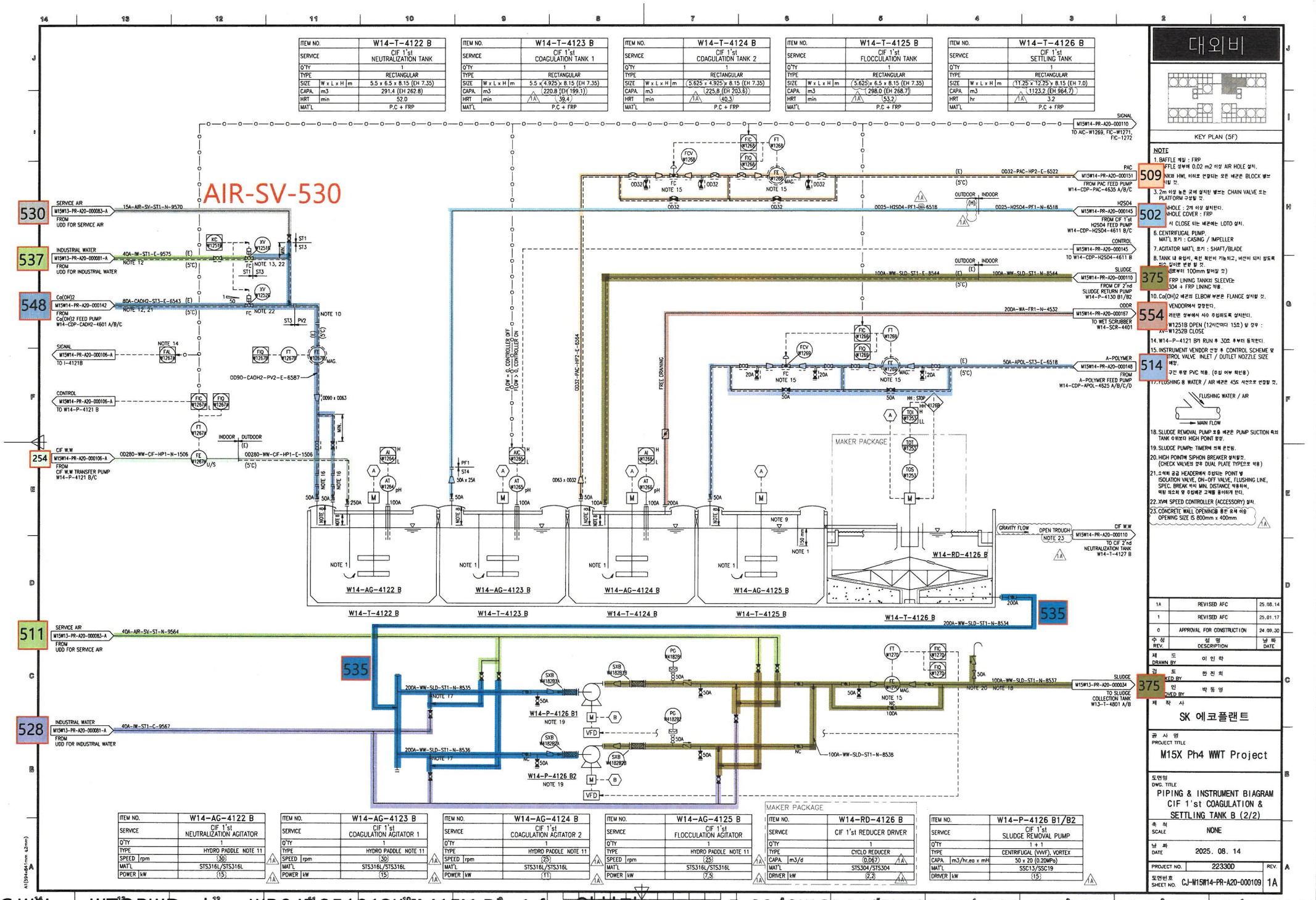

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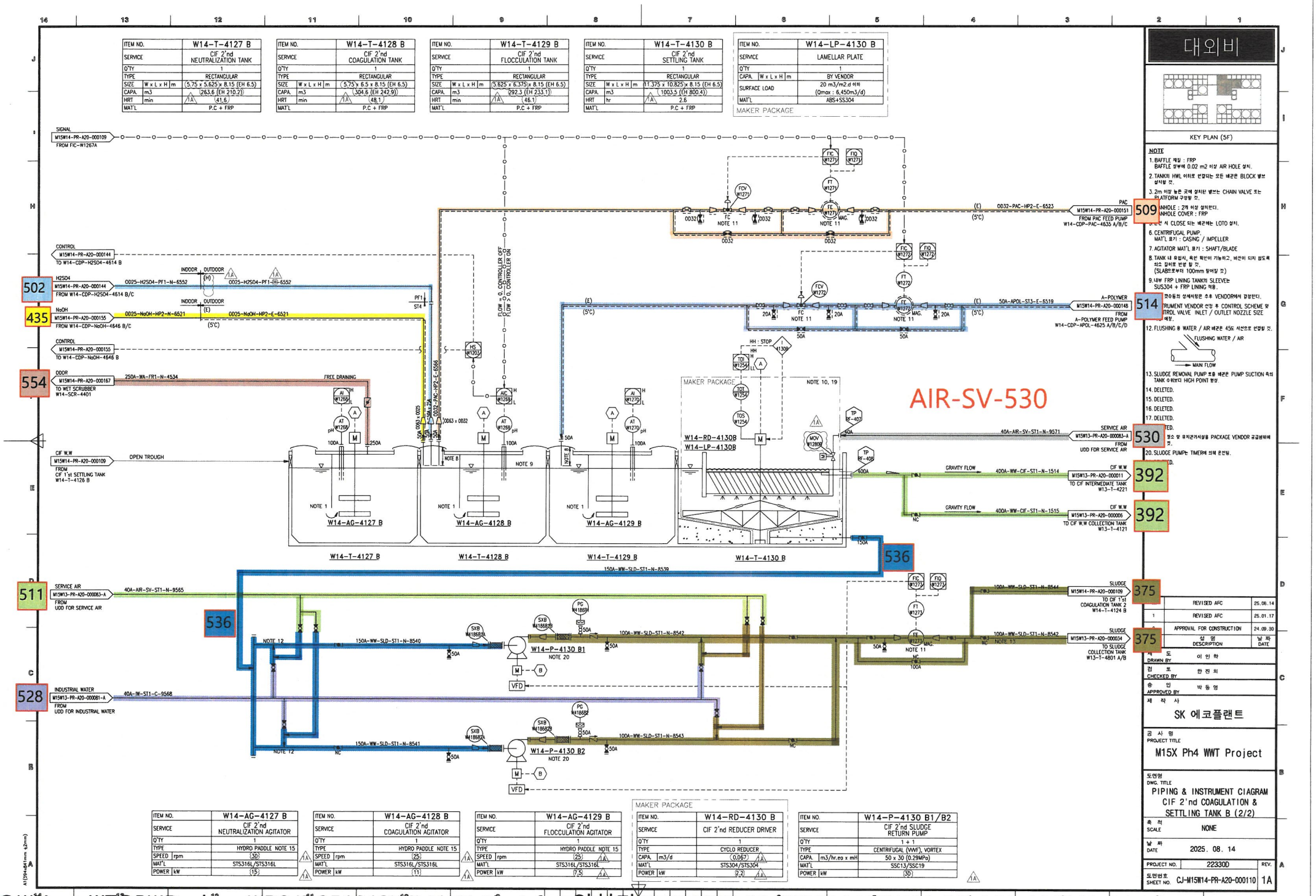

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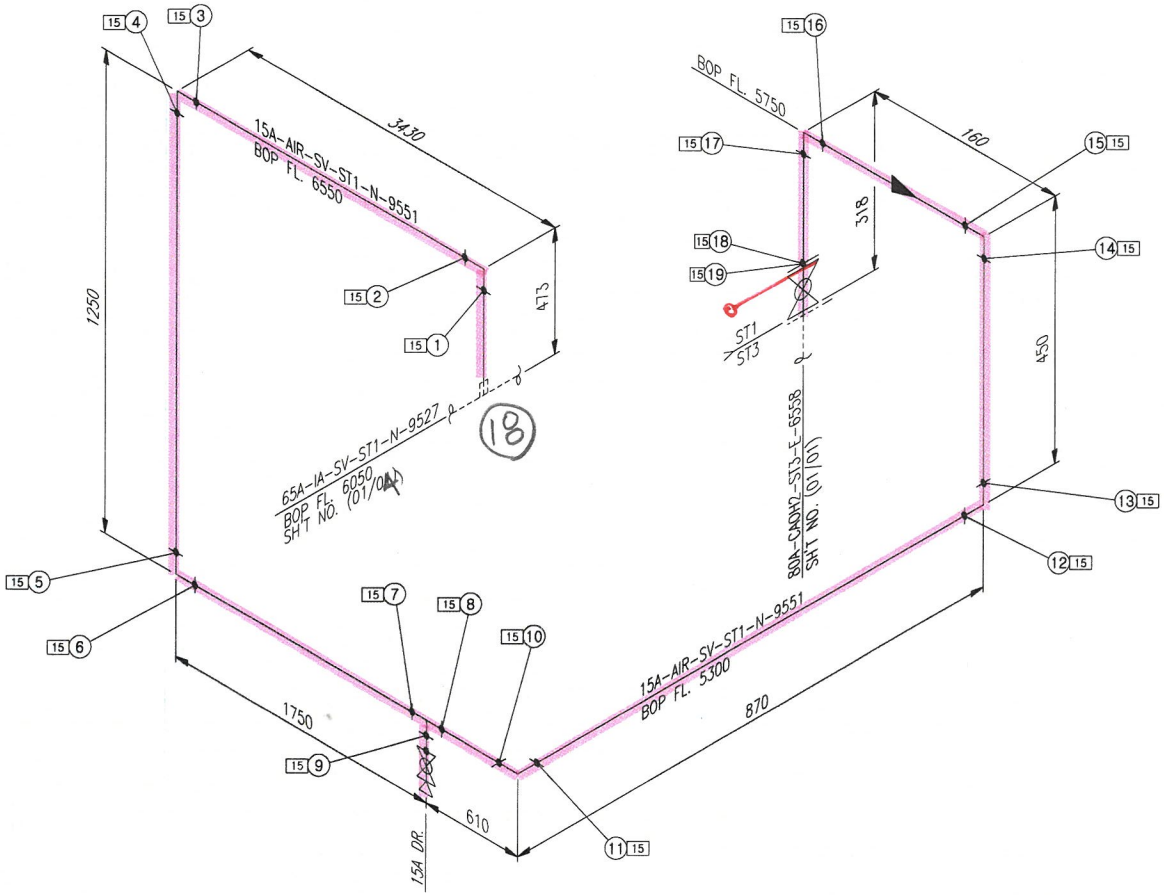


<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	15A	10.2M	PIPE, STS 304TP, ERW, PE - SCH 10S
NP1	15A	1EA	NIPPLE, STS304TP, ERW, TOE - SCH 20S
E90	15A	7EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
T	15A	1EA	TEE EQUAL, STSF304, SW - SCH 10S
VB	15A	1EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
VB1	15A	1EA	BALL VALVE, KS 10K, SCR'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
PLG	15A	1EA	PLUG ROUND HEAD, STSF304, SCR'D
F	15A	1EA	FLG SO, KS 10K, RF, STS304
G	15A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	15A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXS * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (1)

INSULATION	CODE	N
	THICK.	-

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	10	60
OPERATING	7.5	AMB

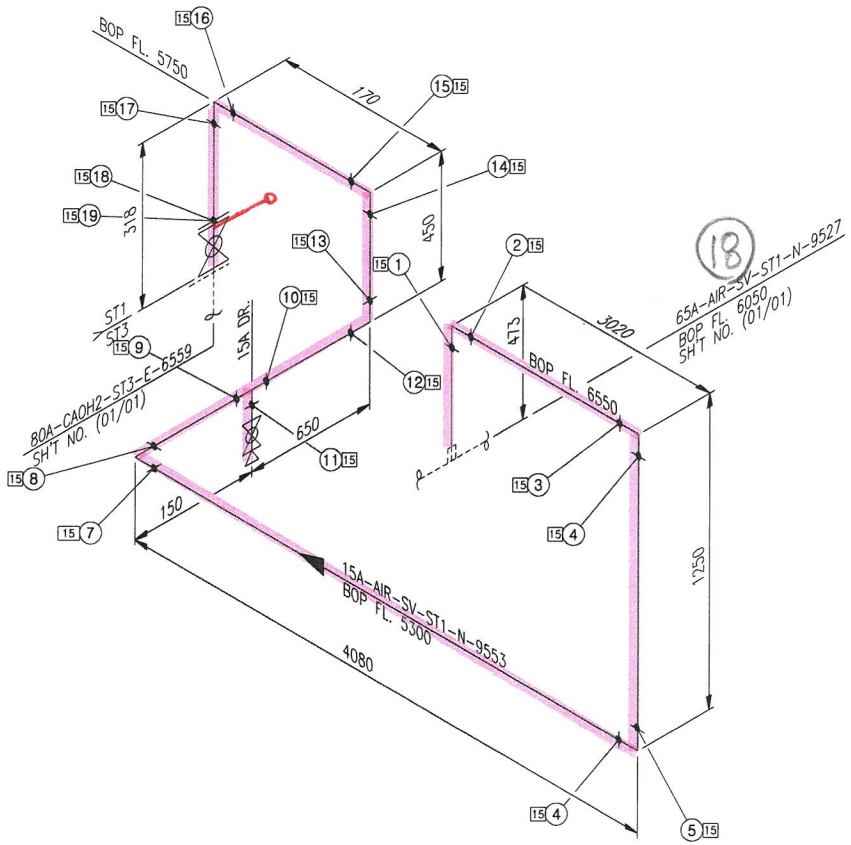
AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10008-B
REF. P & ID CJ-M15W13-PR-A20-000083

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 15A-AIR-SV-ST1-N-9551

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	15A	10.8M	PIPE, STS 304TP, ERW, PE - SCH 10S
NP1	15A	1EA	NIPPLE, STS304TP, ERW, TOE - SCH 20S
E90	15A	7EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
T	15A	1EA	TEE EQUAL, STSF304, SW - SCH 10S
VB	15A	1EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
VB1	15A	1EA	BALL VALVE, KS 10K, SCR'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
PLG	15A	1EA	PLUG ROUND HEAD, STSF304, SCR'D
F	15A	1EA	FLG SO, KS 10K, RF, STS304
G	15A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	15A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	10	60
			OPERATING	7.5	AMB

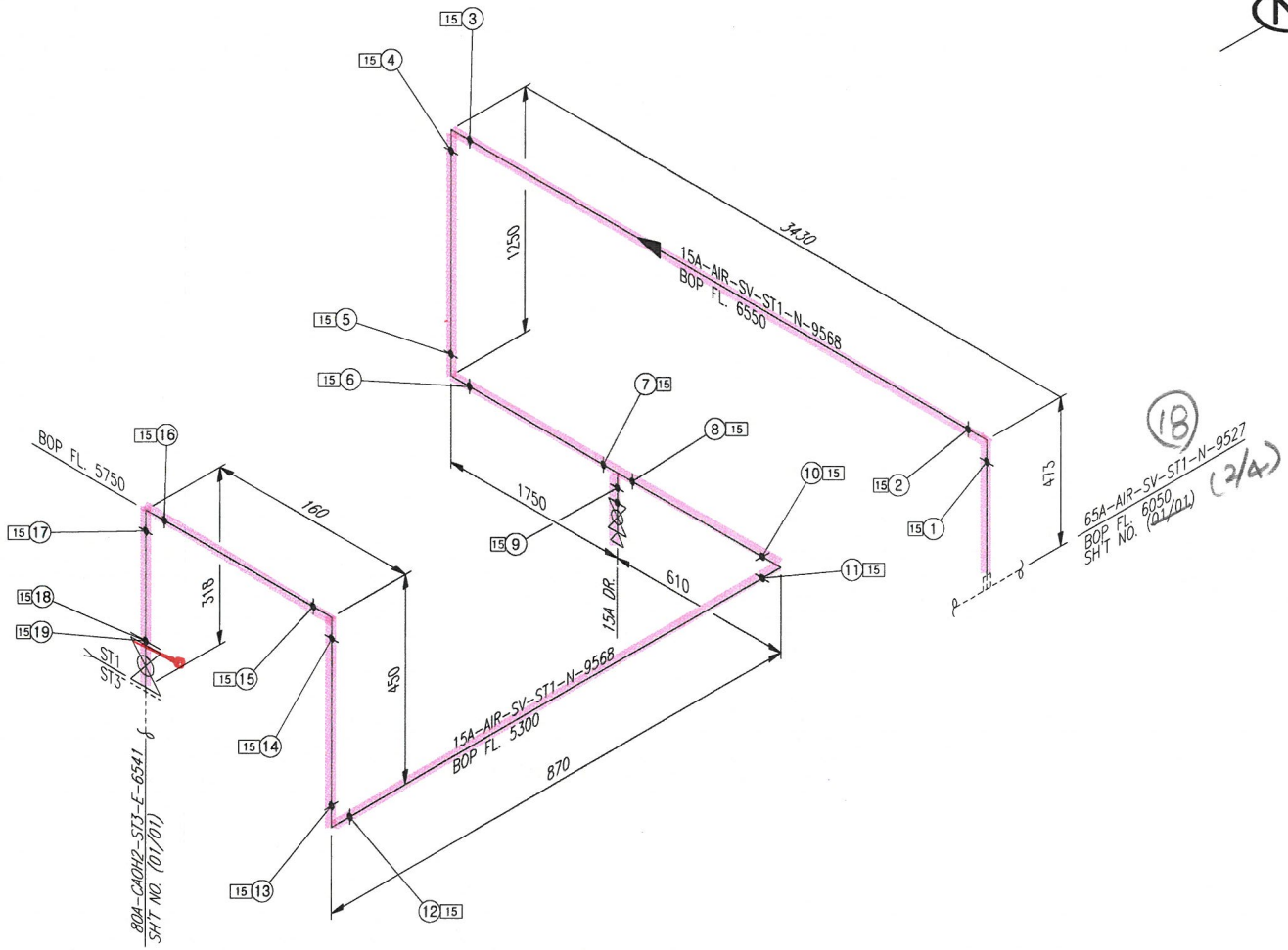
AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10009-B
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동원

CONTRACTOR SK ecoplant
PROJECT NAME M15X Ph4 Project
COMMODITY AIR-SV
JOB NO. 22330D
LINE NO. 15A-AIR-SV-ST1-N-9553
SH'T NO. 01 / 01
REV. 0

CLIENT SK hynix

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	15A	9.0M	PIPE, STS 304TP, ERW, PE – SCH 10S
NP1	15A	1EA	NIPPLE, STS304TP, ERW, TOE – SCH 20S
E90	15A	7EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
T	15A	1EA	TEE EQUAL, STSF304, SW – SCH 10S
VB	15A	1EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
VB1	15A	1EA	BALL VALVE, KS 10K, SCR'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
PLG	15A	1EA	PLUG ROUND HEAD, STSF304, SCR'D
F	15A	1EA	FLG SO, KS 10K, RF, STS304
G	15A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	15A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (3)

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	–	DESIGN	10	60
			OPERATING	7.5	AMB

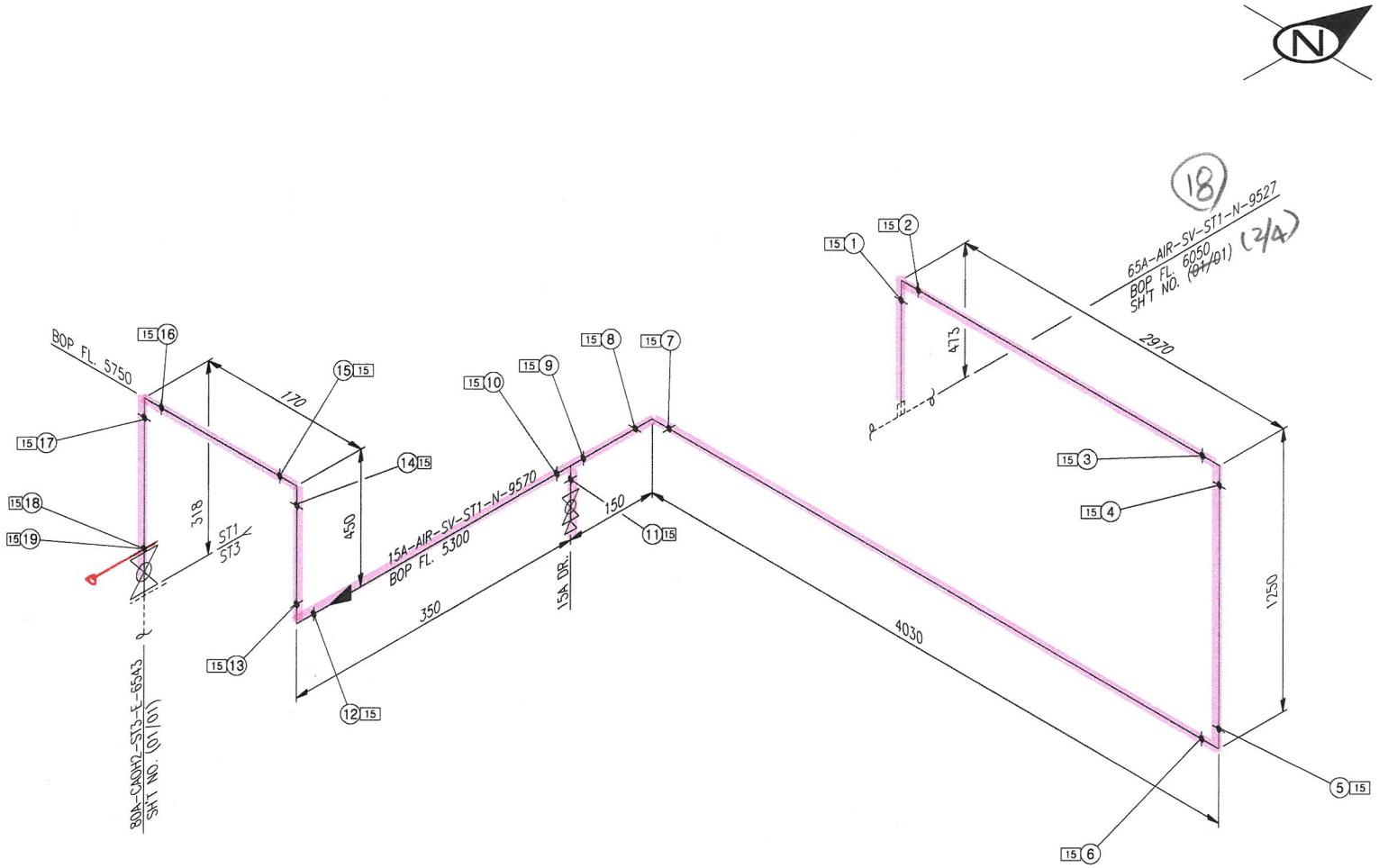
AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10002-B
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 15A-AIR-SV-ST1-N-9568

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	15A	9.9M	PIPE, STS 304TP, ERW, PE - SCH 10S
NP1	15A	1EA	NIPPLE, STS304TP, ERW, TOE - SCH 20S
E90	15A	7EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
T	15A	1EA	TEE EQUAL, STSF304, SW - SCH 10S
VB	15A	1EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
VB1	15A	1EA	BALL VALVE, KS 10K, SCR'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
PLG	15A	1EA	PLUG ROUND HEAD, STSF304, SCR'D
F	15A	1EA	FLG SO, KS 10K, RF, STS304
G	15A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	15A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	10	60
			OPERATING	7.5	AMB

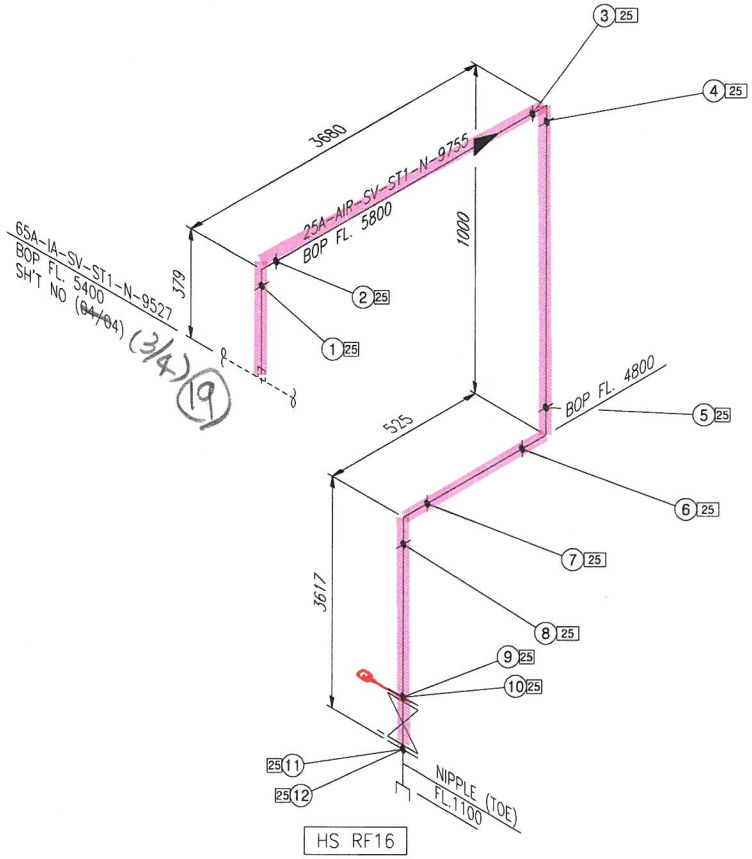
AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10002-B
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 15A-AIR-SV-ST1-N-9570

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	25A	9.2M	PIPE, STS 304TP, ERW, PE - SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
E90	25A	4EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(MALE)

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (5)

INSULATION	CODE	N
	THICK.	-

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	10	60
OPERATING	7.5	AMB

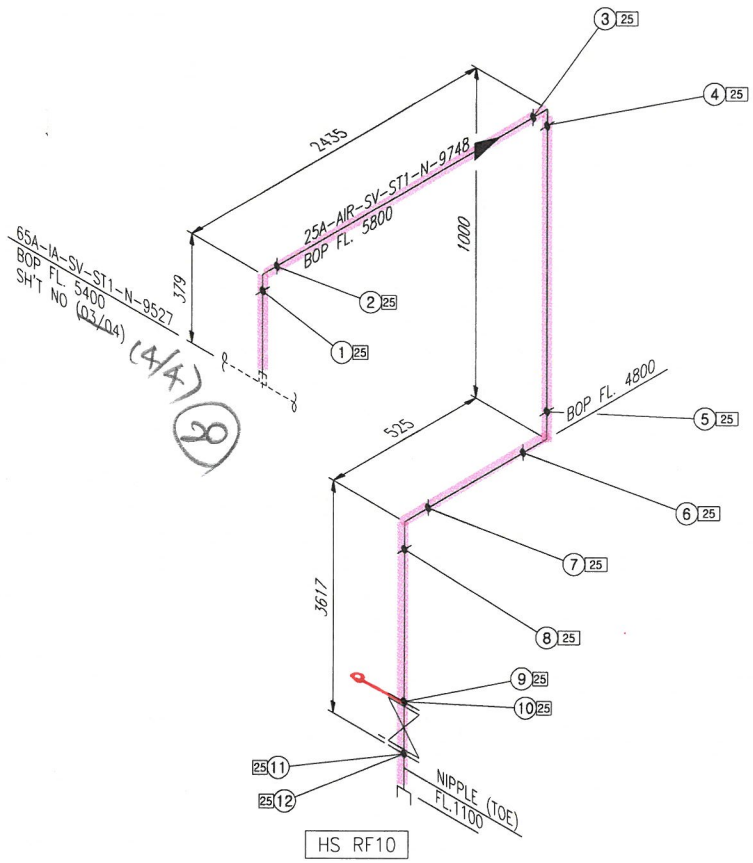
AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10008-B
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 25A-AIR-SV-ST1-N-9755

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	25A	8.0M	PIPE, STS 304TP, ERW, PE - SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
E90	25A	4EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(MALE)

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	10	60
			OPERATING	7.5	AMB

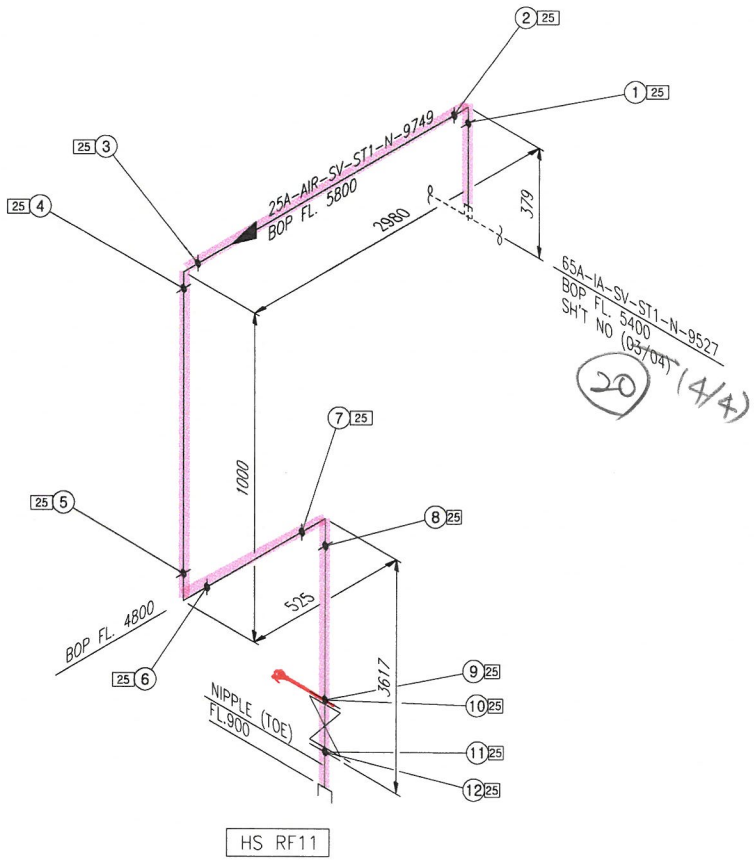
AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10001-B
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 25A-AIR-SV-ST1-N-9748

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	25A	8.5M	PIPE, STS 304TP, ERW, PE - SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
E90	25A	4EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(MALE)

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	10	60
			OPERATING	7.5	AMB

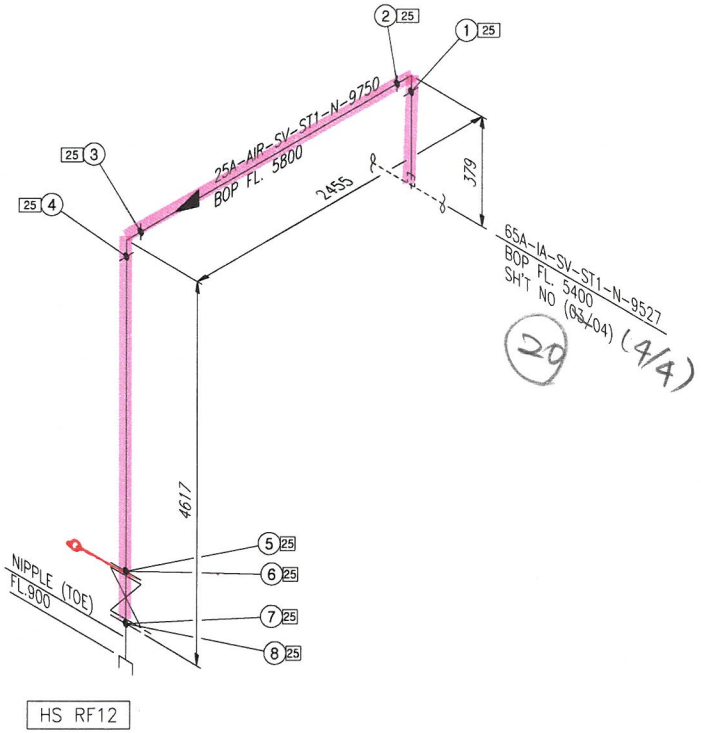
AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10002-B
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 25A-AIR-SV-ST1-N-9749
PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

CLIENT SK hynix

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	25A	7.8M	PIPE, STS 304TP, ERW, PE - SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
E90	25A	2EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(MALE)

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

INSULATION	CODE	N
	THICK.	-

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	10	60
OPERATING	7.5	AMB

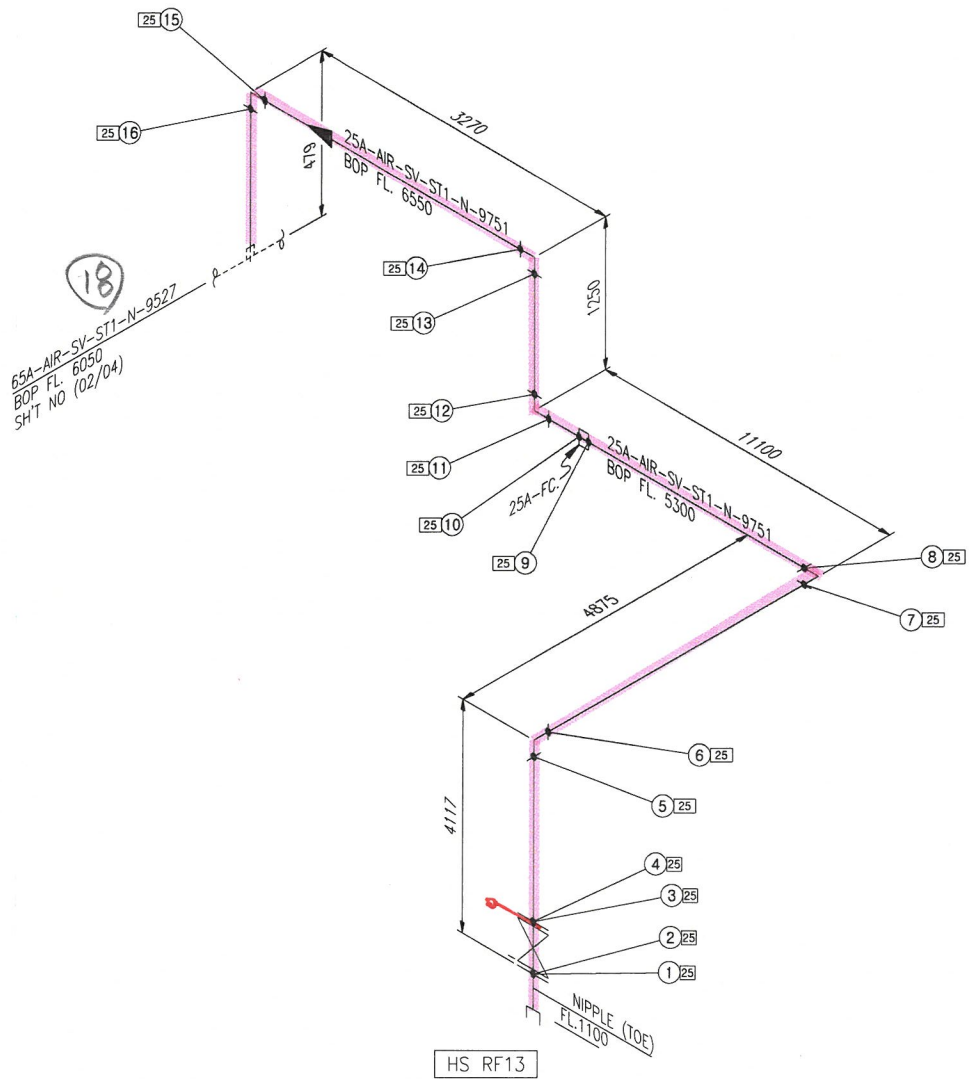
AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10003-B
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 25A-AIR-SV-ST1-N-9750

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	25A	15.4M	PIPE, STS 304TP, ERW, PE - SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
E90	25A	5EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
FC	25A	1EA	COUPLING FULL, STSF304, SW - SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(MALE)

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (9)

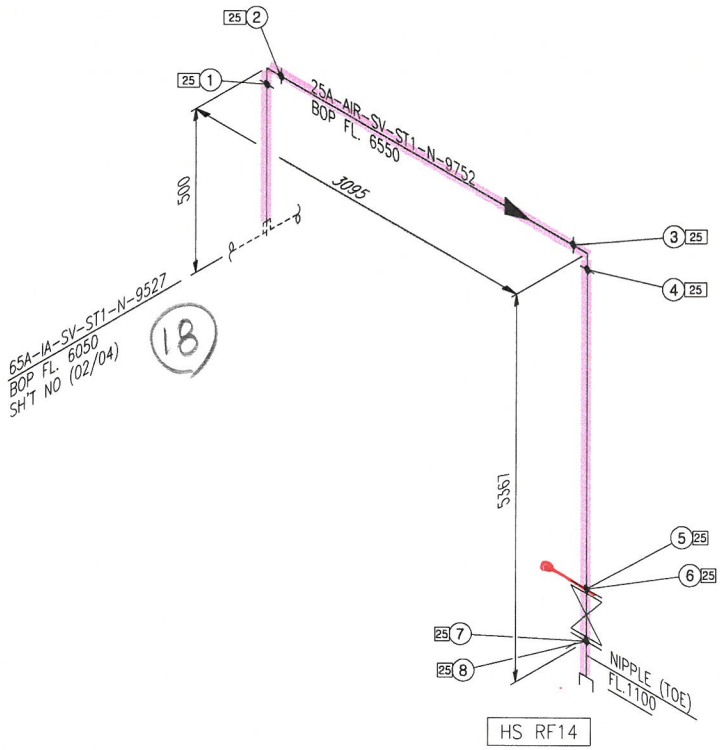
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	10	60
			OPERATING	7.5	AMB

AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10008-B
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 25A-AIR-SV-ST1-N-9751
PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	25A	5.6M	PIPE, STS 304TP, ERW, PE - SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
E90	25A	2EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(MALE)

<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (10)

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	10	60
			OPERATING	7.5	AMB

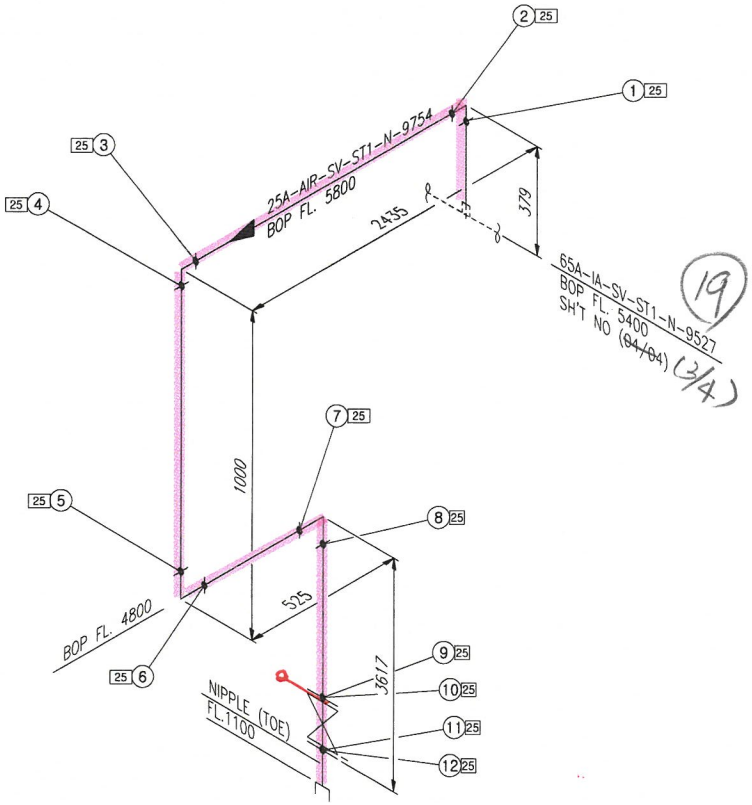
AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10006-B
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 25A-AIR-SV-ST1-N-9752

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



HS RF15

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	25A	8.0M	PIPE, STS 304TP, ERW, PE - SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
E90	25A	4EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(MALE)

<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	10	60
			OPERATING	7.5	AMB

AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10007-B
REF. P & ID CJ-M15W13-PR-A20-000083-A

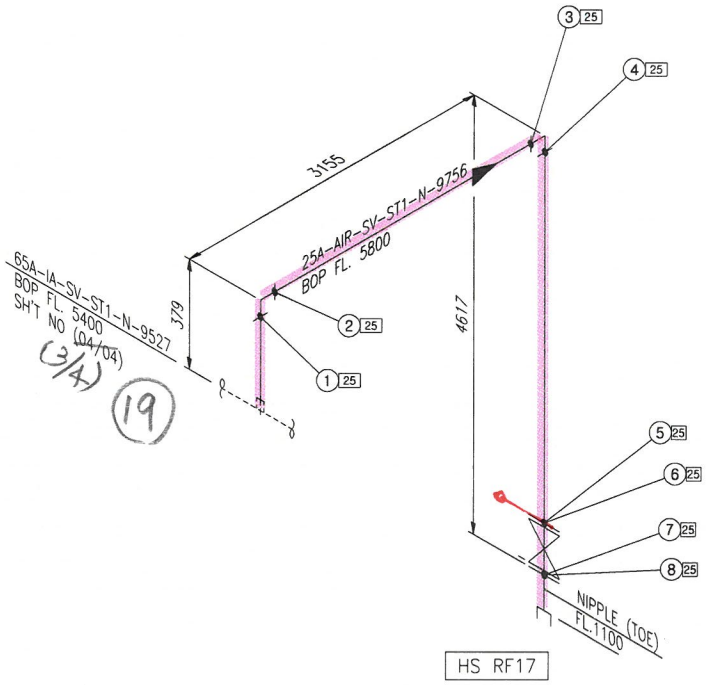
REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CLIENT SK hynix (11)

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 25A-AIR-SV-ST1-N-9754

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	25A	8.2M	PIPE, STS 304TP, ERW, PE - SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
E90	25A	2EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(MALE)

<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (12)

INSULATION	CODE	N
	THICK.	-

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	10	60
OPERATING	7.5	AMB

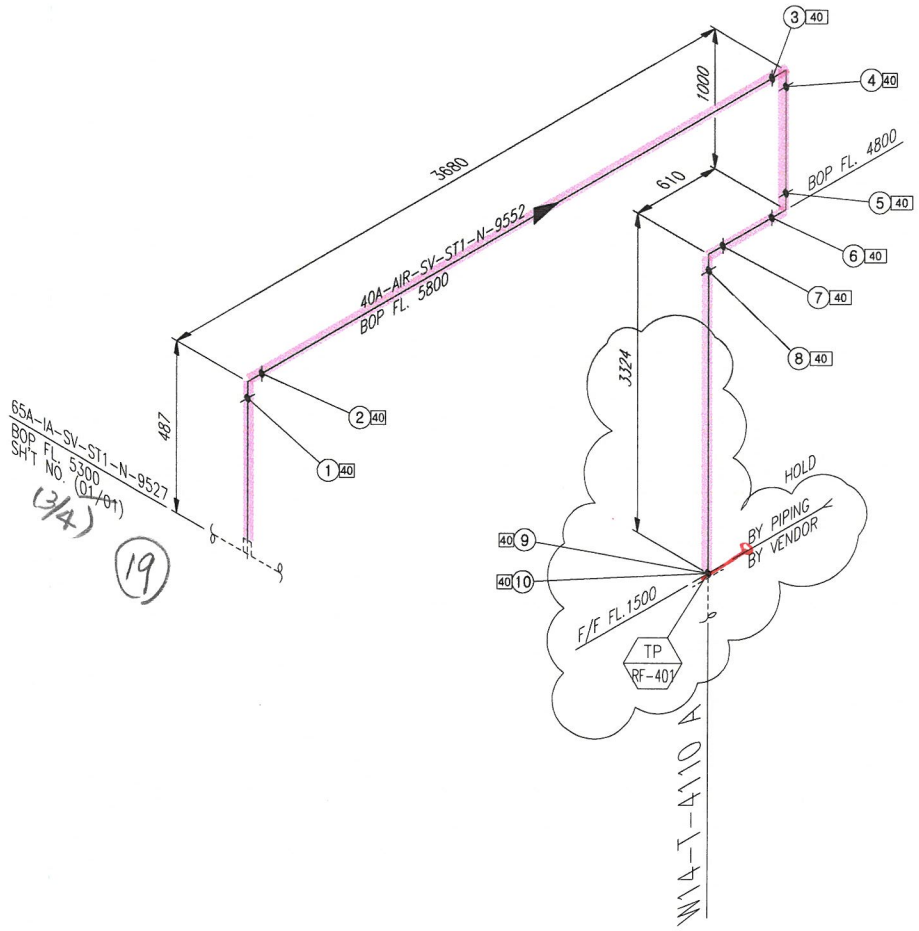
AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10009-B
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 25A-AIR-SV-ST1-N-9756

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	9.2M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	40A	4EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
F	40A	1EA	FLG SO, KS 10K, RF, STS304
G	40A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (13)

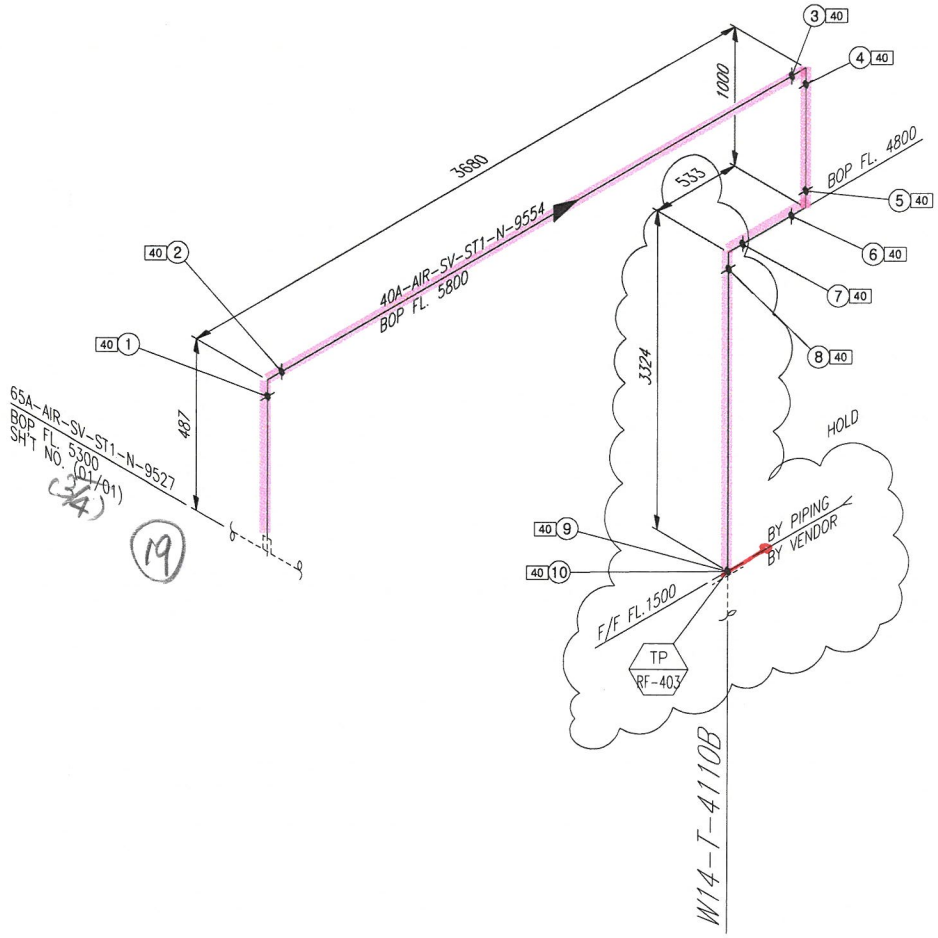
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	10	60
			OPERATING	7.5	AMB

AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10007-B
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 40A-AIR-SV-ST1-N-9552
PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	9.2M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	40A	4EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
F	40A	1EA	FLG SO, KS 10K, RF, STS304
G	40A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	10	60
			OPERATING	7.5	AMB

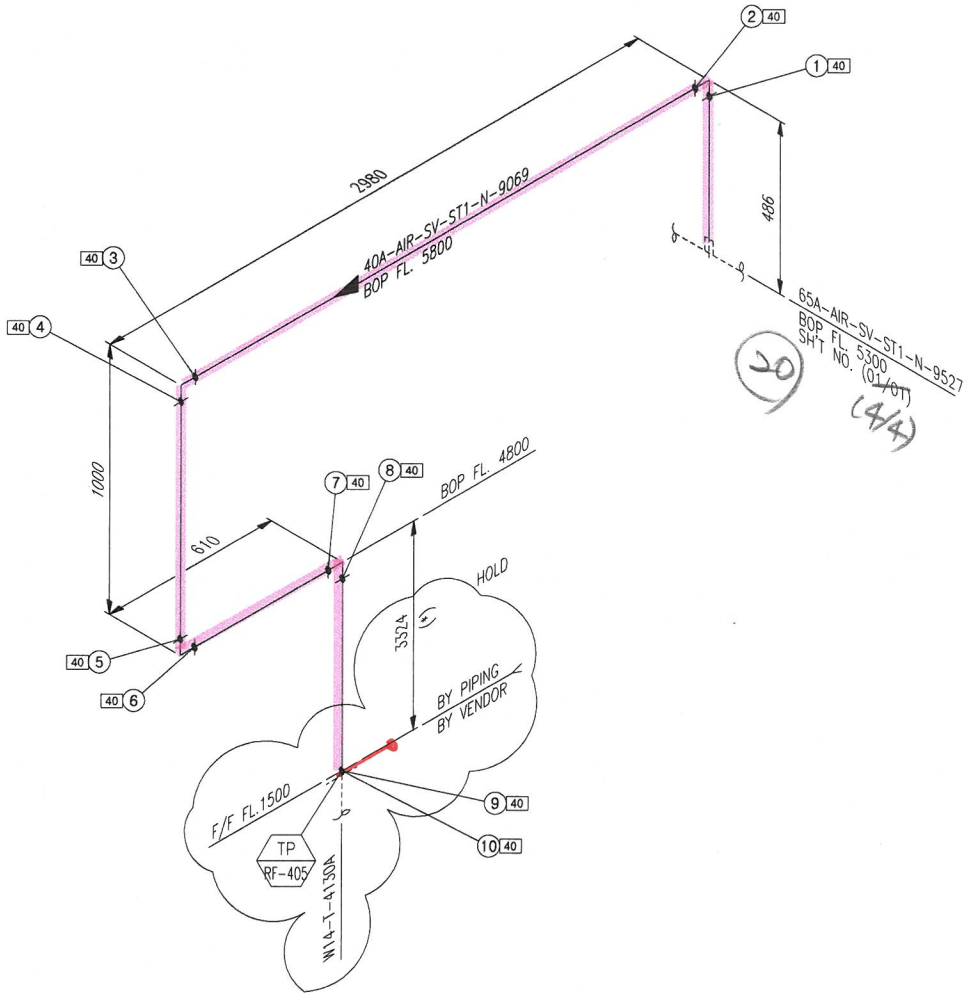
AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10008-B
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 40A-AIR-SV-ST1-N-9554

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	8.5M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	40A	4EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
F	40A	1EA	FLG SO, KS 10K, RF, STS304
G	40A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBL0/VB1/VB1L0/VB2/VB2L0 * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDL0 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (15)

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	10	60
			OPERATING	7.5	AMB

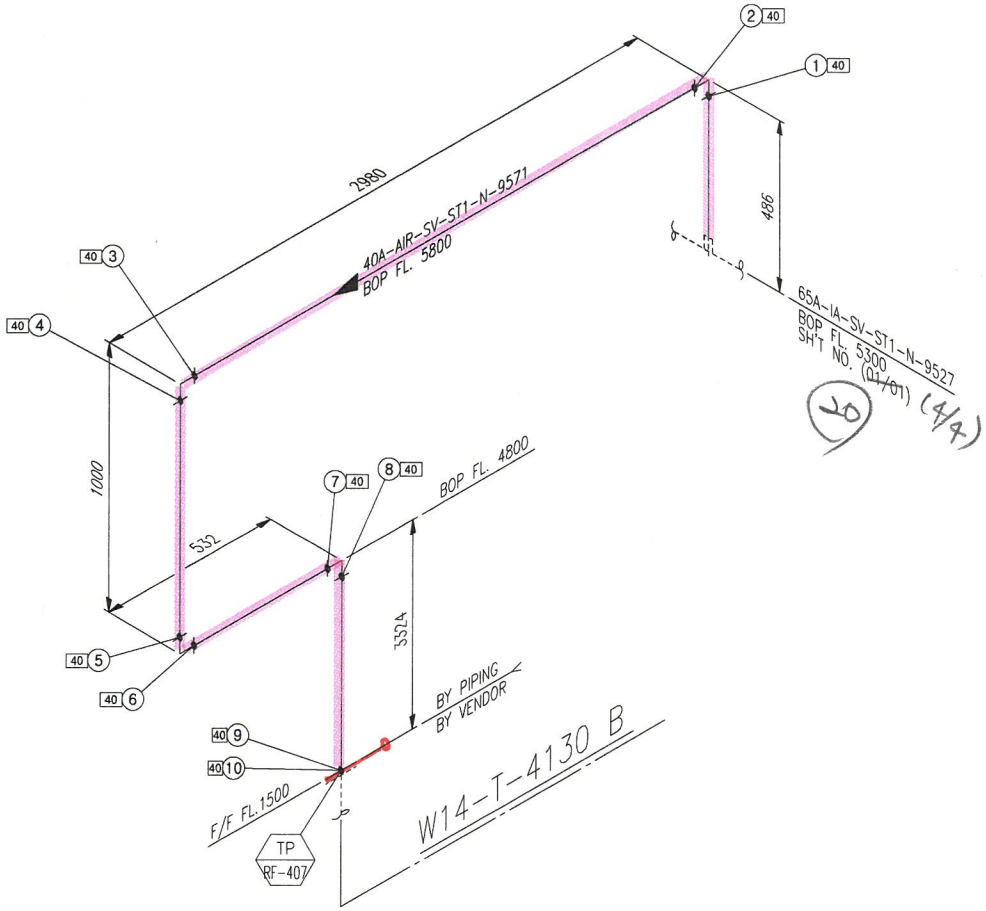
AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10001-B
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래증	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 40A-AIR-SV-ST1-N-9569

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	8.5M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	40A	4EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
F	40A	1EA	FLG SO, KS 10K, RF, STS304
G	40A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (16)

INSULATION	CODE	N
	THICK.	-

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	10	60
OPERATING	7.5	AMB

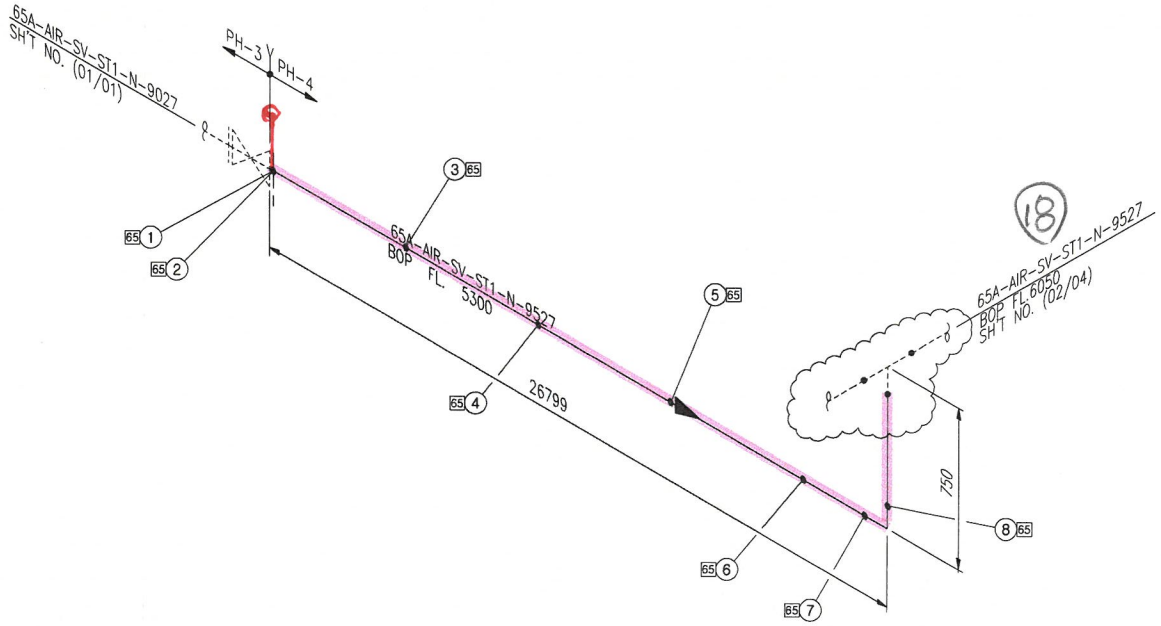
AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10002-B
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 40A-AIR-SV-ST1-N-9571

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	65A	27.5M	PIPE, STS 304TP, ERW, BE - SCH 10S
E90	65A	1EA	ELBOW 90 DEG LR, STS304, ERW, BE - SCH 10S
F	65A	1EA	FLG SO, KS 10K, RF, STS304
G	65A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	65A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (17)

INSULATION	CODE	N
	THICK.	-

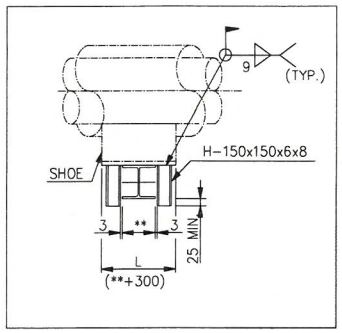
CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	10	60
OPERATING	7.5	AMB

AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10005-B
REF. P & ID CJ-M15W13-PR-A20-000083-A

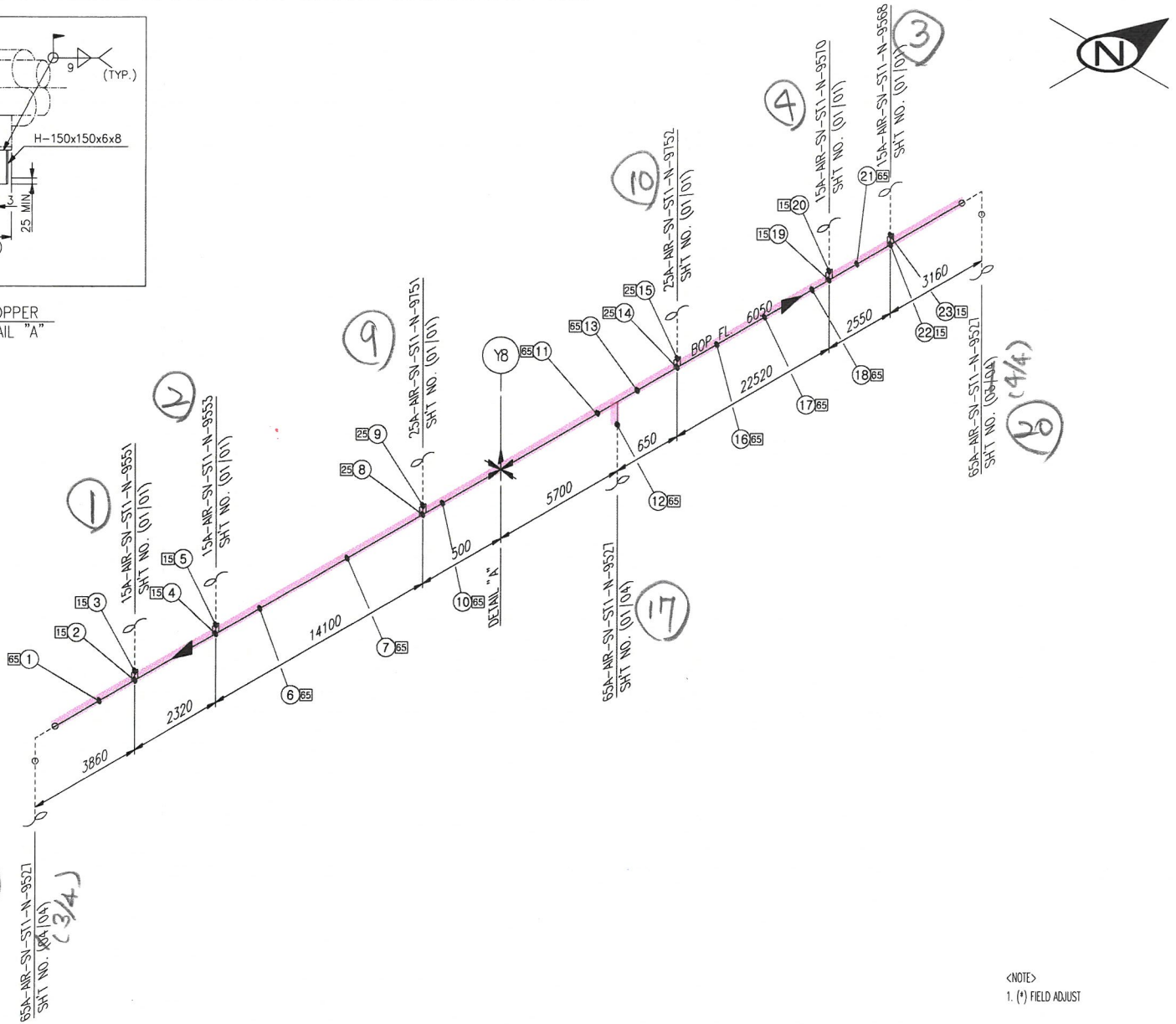
REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 65A-AIR-SV-ST1-N-9527
PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 04
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



STOPPER
DETAIL "A"



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	65A	55.4M	PIPE, STS 304TP, ERW, BE - SCH 10S
T	65A	1EA	TEE EQUAL, STSJ304, ERW, BE - SCH 10S
HF	15A	4EA	COUPLING HALF, STSF304, SW - SCH 10S
HF	25A	2EA	COUPLING HALF, STSF304, SW - SCH 10S

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

18

INSULATION	CODE	N
	THICK.	-

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	10	60
OPERATING	7.5	AMB

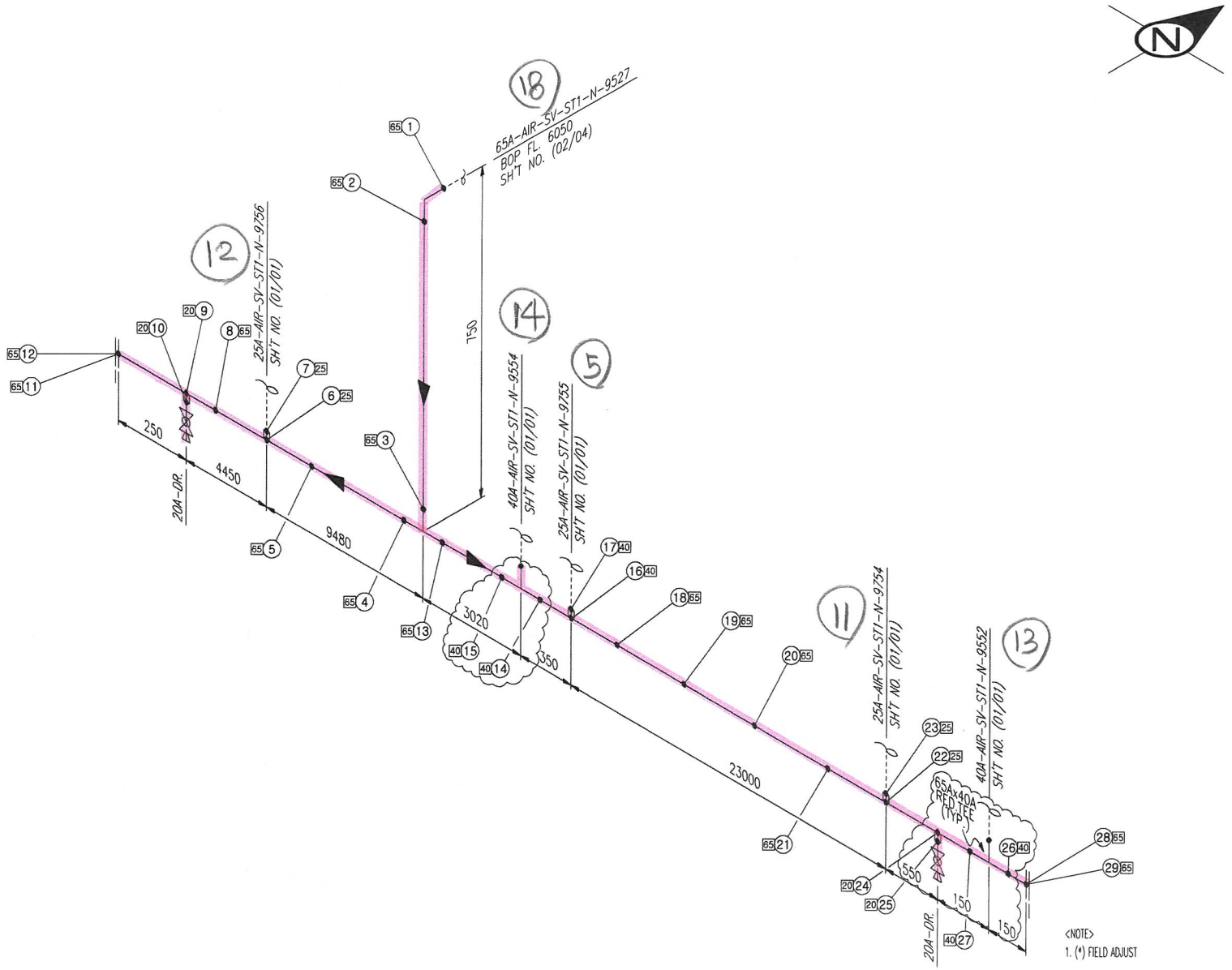
AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10005-B
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 65A-AIR-SV-ST1-N-9527

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 02 / 04
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	65A	42.2M	PIPE, STS 304TP, ERW, BE – SCH 10S
E90	65A	1EA	ELBOW 90 DEG LR, STS304, ERW, BE – SCH 10S
T	65A	1EA	TEE EQUAL, STS304, ERW, BE – SCH 10S
HF	20A	2EA	COUPLING HALF, STSF304, SW – SCH 10S
HF	25A	3EA	COUPLING HALF, STSF304, SW – SCH 10S
HF	40A	2EA	COUPLING HALF, STSF304, SW – SCH 10S
NP1	20A	2EA	NIPPLE, STS 304TP, ERW, TOE – SCH 40S
PLG	20A	2EA	PLUG ROUND HEAD, STSF304, SCR'D
VB1	20A	2EA	BALL VALVE, KS 10K, SCR'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
F	65A	2EA	FLG SO, KS 10K, RF, STS304
FB	65A	2EA	FLG BLIND, KS 10K, RF, STS304
G	65A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	65A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
TR	65Ax40A	2EA	TEE RED., STS304, ERW, BE – SCH 10S

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

19

INSULATION	CODE	N
	THICK.	-

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	10	60
OPERATING	7.5	AMB

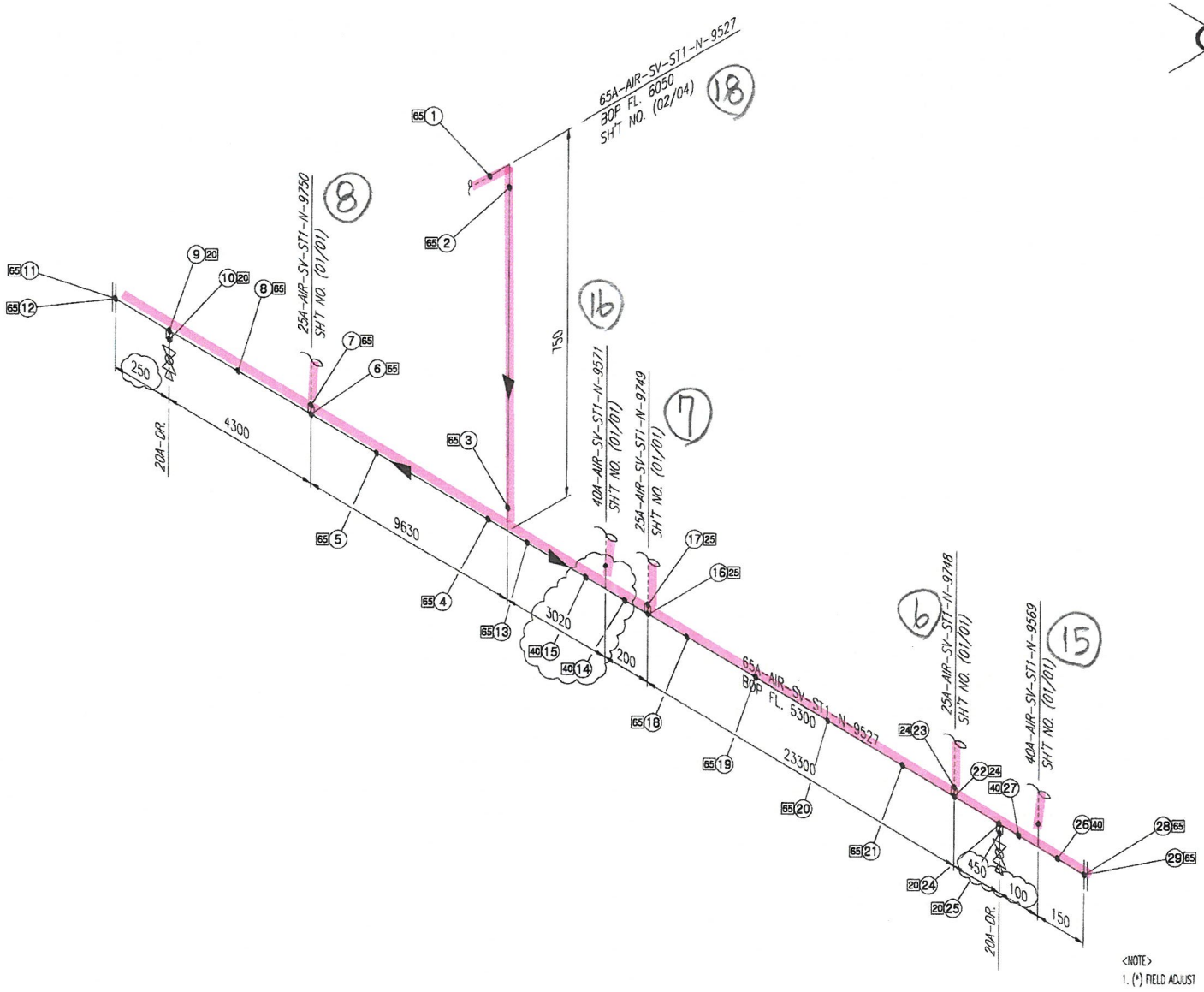
AREA W14
UNIT RFA
REF. PLAN DWG. CJ-M15W14-PI-A30-R10002-B
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 65A-AIR-SV-ST1-N-9527

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 03 / 04
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	65A	42.2M	PIPE, STS 304TP, ERW, BE - SCH 10S
E90	65A	1EA	ELBOW 90 DEG LR, STS304, ERW, BE - SCH 10S
T	65A	1EA	TEE EQUAL, STS304, ERW, BE - SCH 10S
HF	20A	2EA	COUPLING HALF, STSF304, SW - SCH 10S
HF	25A	3EA	COUPLING HALF, STSF304, SW - SCH 10S
HF	40A	2EA	COUPLING HALF, STSF304, SW - SCH 10S
NP1	20A	2EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
PLG	20A	2EA	PLUG ROUND HEAD, STSF304, SCR'D
VB1	20A	2EA	BALL VALVE, KS 10K, SCR'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
F	65A	2EA	FLG SO, KS 10K, RF, STS304
FB	65A	2EA	FLG BLIND, KS 10K, RF, STS304
G	65A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	65A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
TR	65Ax40A	2EA	RED. TEE, STS304, ERW, BE - SCH 10S

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBL0/VB1/VB1L0/VB2/VB2L0 * BUTTERFLY VA. : VF/VF1 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VCS/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDL0 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL, PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14					
	THICK.	-	DESIGN	10	60	UNIT RFA					
			OPERATING	7.5	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10008-B					
						REF. P & ID CJ-M15W13-PR-A20-000083-A					
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
							0	25.01.24	APPROVED FOR CONSTRUCTION	황태종	안남후	박동영

CLIENT SK hynix 20

CONTRACTOR SK ecoplant

PROJECT NAME M15X Ph4 Project

COMMODITY AIR-SV

JOB NO. 22330D

LINE NO. 65A-AIR-SV-ST1-N-9527

SH'T NO. 04 / 04

REV. 0

PUNCH LIST

Project No : 22330D

Project Title : M15X Ph3 Project

Package No :

Date :

S/N	Drawing No	Description	Category A B C	Punch Clear Status			
				작성자		확인자	
				Name	Sign	Name	Sign
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							

Remarks

A : Before Hydro Test

B : Before M/C

C : Before Commissioning